

4.28.14 Number Representation: GATE CSE 2000 | Question: 1.6



The number 43 in $2's$ complement representation is ¹

- A. 01010101 B. 11010101 C. 00101011 D. 10101011 ²

gatecse-2000 digital-logic number-representation easy ³

Answer key

4.28.15 Number Representation: GATE CSE 2000 | Question: 2.14



Consider the values of $A = 2.0 \times 10^{30}$, $B = -2.0 \times 10^{30}$, $C = 1.0$, and the sequence ⁴

$X := A + B$ $Y := A + C$ ⁵
 $X := X + C$ $Y := Y + B$

executed on a computer where floating point numbers are represented with 32 bits. The values for X and Y will be ⁶

- A. $X = 1.0, Y = 1.0$ B. $X = 1.0, Y = 0.0$ ⁷
 C. $X = 0.0, Y = 1.0$ D. $X = 0.0, Y = 0.0$

gatecse-2000 digital-logic number-representation normal ⁸

Answer key

4.28.16 Number Representation: GATE CSE 2001 | Question: 2.10



The $2's$ complement representation of $(-539)_{10}$ in hexadecimal is ¹⁰

- A. ABE B. DBC C. $DE5$ D. $9E7$ ¹¹

gatecse-2001 digital-logic number-representation easy ¹²

Answer key

4.28.17 Number Representation: GATE CSE 2002 | Question: 1.14



The decimal value 0.25 ¹³

- A. is equivalent to the binary value 0.1 ¹⁴
 B. is equivalent to the binary value 0.01
 C. is equivalent to the binary value 0.00111
 D. cannot be represented precisely in binary

gatecse-2002 digital-logic number-representation easy ¹⁵

Answer key

4.28.18 Number Representation: GATE CSE 2002 | Question: 1.15



The $2's$ complement representation of the decimal value -15 is ¹⁷

- A. 1111 B. 11111 C. 111111 D. 10001 ¹⁸

gatecse-2002 digital-logic number-representation easy ¹⁹

Answer key ²⁰

4.28.19 Number Representation: GATE CSE 2002 | Question: 1.16



Sign extension is a step in

- A. floating point multiplication B. signed 16 bit integer addition
 C. arithmetic left shift D. converting a signed integer from one size to another

gatecse-2002 digital-logic easy number-representation

Answer key

4.28.20 Number Representation: GATE CSE 2002 | Question: 1.21



In 2^s complement addition, overflow ¹

- A. is flagged whenever there is carry from sign bit addition ²
- B. cannot occur when a positive value is added to a negative value
- C. is flagged when the carries from sign bit and previous bit match
- D. None of the above

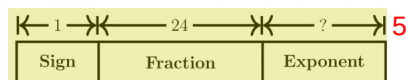
gatecse-2002 digital-logic number-representation normal ³

Answer key

4.28.21 Number Representation: GATE CSE 2002 | Question: 9



Consider the following 32-bit floating-point representation scheme as shown in the format below. A value is specified by 3 fields, a one bit sign field (with 0 for positive and 1 for negative values), a 24 bit fraction field (with the binary point is at the left end of the fraction bits), and a 7 bit exponent field (in excess-64 signed integer representation, with 16 is the base of exponentiation). The sign bit is the most significant bit. ⁴



- A. It is required to represent the decimal value -7.5 as a normalized floating point number in the given format. Derive the values of the various fields. Express your final answer in the hexadecimal. ⁶
- B. What is the largest value that can be represented using this format? Express your answer as the nearest power of 10.

gatecse-2002 digital-logic number-representation normal descriptive ⁷

Answer key

4.28.22 Number Representation: GATE CSE 2003 | Question: 9



Assuming all numbers are in 2^s complement representation, which of the following numbers is divisible by 8? ⁸

- A. 11100111
- B. 11100100
- C. 11010111
- D. 11011011 ⁹

gatecse-2003 digital-logic number-representation normal ¹⁰

Answer key

4.28.23 Number Representation: GATE CSE 2004 | Question: 19



If 73_x (in base- x number system) is equal to 54_y (in base- y -number system), the possible values of x and y are ¹¹

- A. 8, 16
- B. 10, 12
- C. 9, 13
- D. 8, 11 ¹³

gatecse-2004 digital-logic number-representation easy ¹⁴

Answer key

4.28.24 Number Representation: GATE CSE 2004 | Question: 28



What is the result of evaluating the following two expressions using three-digit floating point arithmetic with rounding?

$$(113. + -111.) + 7.51$$

$$113. + (-111. + 7.51)$$

- A. 9.51 and 10.0 respectively
- B. 10.0 and 9.51 respectively
- C. 9.51 and 9.51 respectively
- D. 10.0 and 10.0 respectively

gatecse-2004 digital-logic number-representation normal

Answer key

4.28.25 Number Representation: GATE CSE 2004 | Question: 66

Let $A = 11111010$ and $B = 00001010$ be two 8-bit 2's complement numbers. Their product in 2's complement is

- A. 11000100 B. 10011100 C. 10100101 D. 11010101 ²

gatecse-2004 digital-logic number-representation easy ³

Answer key

4.28.26 Number Representation: GATE CSE 2005 | Question: 16, ISRO2009-18, ISRO2015-2

The range of integers that can be represented by an n bit 2's complement number system is: ⁵

- A. -2^{n-1} to $(2^{n-1} - 1)$ B. $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$ ⁶
C. -2^{n-1} to 2^{n-1} D. $-(2^{n-1} + 1)$ to $(2^{n-1} - 1)$

gatecse-2005 digital-logic number-representation easy isro2009 isro2015 ⁷

Answer key ⁸

4.28.27 Number Representation: GATE CSE 2005 | Question: 17

The hexadecimal representation of $(657)_8$ is: ¹⁰

- A. 1AF B. D78 C. D71 D. 32F ¹¹

gatecse-2005 digital-logic number-representation easy ¹²

Answer key

4.28.28 Number Representation: GATE CSE 2006 | Question: 39

We consider the addition of two 2's complement numbers $b_{n-1}b_{n-2} \dots b_0$ and $a_{n-1}a_{n-2} \dots a_0$. A binary adder for adding unsigned binary numbers is used to add the two numbers. The sum is denoted by $c_{n-1}c_{n-2} \dots c_0$ and the carry-out by c_{out} . Which one of the following options correctly identifies the overflow condition? ¹³

- A. $c_{out} (\overline{a_{n-1} \oplus b_{n-1}})$ B. $a_{n-1}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}}\overline{b_{n-1}}c_{n-1}$ ¹⁵
C. $c_{out} \oplus c_{n-1}$ D. $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

gatecse-2006 digital-logic number-representation normal ¹⁶

Answer key ¹⁷

4.28.29 Number Representation: GATE CSE 2008 | Question: 6

Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are ¹⁹

- A. decimal 10 B. decimal 11 C. decimal 10 and 11 D. any value > 2 ²⁰

gatecse-2008 digital-logic number-representation normal ²¹

Answer key

4.28.30 Number Representation: GATE CSE 2009 | Question: 5, ISRO2017-57

$(1217)_8$ is equivalent to ²³

- A. $(1217)_{16}$ B. $(028F)_{16}$ C. $(2297)_{10}$ D. $(0B17)_{16}$ ²⁴

gatecse-2009 digital-logic number-representation isro2017 ²⁵

Answer key

4.28.31 Number Representation: GATE CSE 2010 | Question: 8

P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement

representation of $8 \times P$ is **1**

- A. $(C3D8)_{16}$ B. $(187B)_{16}$ C. $(F878)_{16}$ D. $(987B)_{16}$ **2**

gatecse-2010 digital-logic number-representation normal **3**

Answer key

4.28.32 Number Representation: GATE CSE 2013 | Question: 4

The smallest integer that can be represented by an 8-bit number in 2 's complement form is **4**

- A. -256 B. -128 C. -127 D. 0 **5**

gatecse-2013 digital-logic number-representation easy **6**

Answer key

4.28.33 Number Representation: GATE CSE 2014 | Set 1 | Question: 8

The base (or radix) of the number system such that the following equation holds is **7**

$$\frac{312}{20} = 13.1$$
 8

gatecse-2014-set1 digital-logic number-representation numerical-answers normal **9**

Answer key **10**

4.28.34 Number Representation: GATE CSE 2014 | Set 2 | Question: 8

Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is **11**

gatecse-2014-set2 digital-logic number-representation numerical-answers normal **12**

Answer key

4.28.35 Number Representation: GATE CSE 2015 | Set 3 | Question: 35

Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solutions is **13**

gatecse-2015-set3 digital-logic number-representation normal numerical-answers **14**

Answer key

4.28.36 Number Representation: GATE CSE 2016 | Set 1 | Question: 07

The 16-bit 2 's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is **15**

gatecse-2016-set1 digital-logic number-representation normal numerical-answers **16**

Answer key

4.28.37 Number Representation: GATE CSE 2016 | Set 2 | Question: 09

Let X be the number of distinct 16-bit integers in 2 's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is **17**

gatecse-2016-set2 digital-logic number-representation normal numerical-answers

Answer key

4.28.38 Number Representation: GATE CSE 2017 | Set 1 | Question: 9

When two 8-bit numbers $A_7 \cdots A_0$ and $B_7 \cdots B_0$ in 2 's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \cdots S_0$ and the carry bits are $C_7 \cdots C_0$. An overflow is said to have occurred if

- A. the carry bit C_7 is **1**

- B. all the carry bits $(C_7, \dots, C_0)$ are 1 ¹
- C. $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1
- D. $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1

gatecse-2017-set1 digital-logic number-representation

Answer key

4.28.39 Number Representation: GATE CSE 2017 | Set 2 | Question: 1

The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA92. The representation of the value of X in octal number system is

- A. 571244 B. 736251 C. 571247 D. 136251 ³

gatecse-2017-set2 digital-logic number-representation

Answer key

4.28.40 Number Representation: GATE CSE 2019 | Question: 22

Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

gatecse-2019 numerical-answers digital-logic number-representation probability one-mark ⁵

Answer key

4.28.41 Number Representation: GATE CSE 2019 | Question: 4

In 16-bit 2's complement representation, the decimal number -28 is: ⁶

- A. 1111 1111 0001 1100 B. 0000 0000 1110 0100 ⁷
- C. 1111 1111 1110 0100 D. 1000 0000 1110 0100

gatecse-2019 digital-logic number-representation one-mark ⁸

Answer key

4.28.42 Number Representation: GATE CSE 2019 | Question: 8

Consider $Z = X - Y$ where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of: ⁹

- A. n bits B. $n - 1$ bits C. $n + 1$ bits D. $n + 2$ bits ¹¹

gatecse-2019 digital-logic number-representation one-mark ¹²

Answer key

4.28.43 Number Representation: GATE CSE 2021 | Set 1 | Question: 6

Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number? ¹³

- A. 15 B. 21 C. D2 D. 528 ¹⁵

gatecse-2021-set1 digital-logic number-representation normal one-mark ¹⁶

Answer key

4.28.44 Number Representation: GATE CSE 2021 | Set 2 | Question: 18

If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is _____

gatecse-2021-set2 numerical-answers digital-logic number-representation easy one-mark

Answer key

4.28.45 Number Representation: GATE CSE 2022 | Question: 8



Let R_1 and R_2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R_1 + R_2$, which one of the following values of R_1 and R_2 gives an arithmetic overflow?

- A. $R_1 = 1011$ and $R_2 = 1110$ B. $R_1 = 1100$ and $R_2 = 1010$
 C. $R_1 = 0011$ and $R_2 = 0100$ D. $R_1 = 1001$ and $R_2 = 1111$

gatecse-2022 digital-logic number-representation one-mark 3

Answer key

4.28.46 Number Representation: GATE CSE 2023 | Question: 22



A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.

gatecse-2023 digital-logic number-representation numerical-answers one-mark 5

Answer key

4.28.47 Number Representation: GATE CSE 2024 | Set 1 | Question: 3



Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B=11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

- A. $A + B$ B. $A - B$ C. $B - A$ D. $2 * B$

gatecse-2024-set1 digital-logic number-representation one-mark 9

Answer key

4.28.48 Number Representation: GATE CSE 2024 | Set 2 | Question: 39



Which of the following is/are EQUAL to 224 in radix - 5 (i.e., base - 5) notation?

- A. 64 in radix -10 B. 100 in radix -8
 C. 50 in radix -16 D. 121 in radix -7

gatecse-2024-set2 digital-logic number-representation easy multiple-selects two-marks 12

Answer key

4.28.49 Number Representation: GATE CSE 2025 | Set 1 | Question: 15



The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6 ?

- A. 1000 1010 in 8-bits B. 1111 1010 in 8-bits
 C. 1000 0000 0000 1010 in 16-bits D. 1111 1111 1111 1010 in 16-bits

gatecse2025-set1 digital-logic number-representation multiple-selects easy one-mark 15

Answer key

4.28.50 Number Representation: GATE CSE 2026 | Set 1 | Question: 12



Consider the 8-bit signed integers X , Y and Z represented using the sign-magnitude form. The binary representations of X and Y are as follows:

$X : 10110100$ $Y : 01001100$

Which of the following operations to compute Z result(s) in an arithmetic overflow?

- A. $Z = X + Y$ B. $Z = X - Y$ C. $Z = -X + Y$ D. $Z = -X - Y$

gatecse-2026-set1 digital-logic number-representation multiple-selects one-mark

Answer key

4.28.51 Number Representation: GATE CSE 2026 | Set 2 | Question: 18



In a system, numbers are represented using 4-bit two's complement form. Consider four numbers $N_1 = 1011$, $N_2 = 1101$, $N_3 = 1010$ and $N_4 = 1001$ in the system. Which of the following operations will result in arithmetic overflow?

- A. $N_1 + N_2$ B. $N_2 + N_3$ C. $N_3 - N_4$ D. $N_1 + N_4$ ²

gatecse-2026-set2 digital-logic number-representation multiple-selects one-mark ³

Answer key

4.28.52 Number Representation: GATE IT 2004 | Question: 42



Using a 4 – *bit 2's* complement arithmetic, which of the following additions will result in an overflow? ⁴

- i. $1100 + 1100$ ⁵
ii. $0011 + 0111$
iii. $1111 + 0111$

- A. i only B. ii only C. iii only D. i and iii only ⁶

gateit-2004 digital-logic number-representation normal ⁷

Answer key

4.28.53 Number Representation: GATE IT 2004 | Question: 43



The number $(123456)_8$ is equivalent to ⁸

- A. $(A72E)_{16}$ and $(22130232)_4$ B. $(A72E)_{16}$ and $(22131122)_4$ ⁹
C. $(A73E)_{16}$ and $(22130232)_4$ D. $(A62E)_{16}$ and $(22120232)_4$

gateit-2004 digital-logic number-representation normal ¹⁰

Answer key ¹¹

4.28.54 Number Representation: GATE IT 2005 | Question: 47



$(34.4)_8 \times (23.4)_8$ evaluates to ¹²

- A. $(1053.6)_8$ B. $(1053.2)_8$ C. $(1024.2)_8$ D. None of these ¹³

gateit-2005 digital-logic number-representation normal ¹⁴

Answer key ¹⁵

4.28.55 Number Representation: GATE IT 2006 | Question: 7, ISRO2009-41



The addition of 4 – *bit*, two's complement, binary numbers 1101 and 0100 results in ¹⁷

- A. 0001 and an overflow B. 1001 and no overflow ¹⁸
C. 0001 and no overflow D. 1001 and an overflow

gateit-2006 digital-logic number-representation normal isro2009 ¹⁹

Answer key

4.28.56 Number Representation: GATE IT 2007 | Question: 42



$(C012.25)_H - (10111001110.101)_B =$

- A. $(135103.412)_O$ B. $(564411.412)_O$
C. $(564411.205)_O$ D. $(135103.205)_O$

gateit-2007 digital-logic number-representation normal

Answer key

4.28.57 Number Representation: GATE IT 2008 | Question: 15



A processor that has the carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this

addition operation, the status of the carry, overflow and sign flags, respectively will be: 1

- A. 1,1,0 B. 1,0,0 C. 0,1,0 D. 1,0,1 2

gateit-2008 digital-logic number-representation normal co-and-architecture 3

Answer key

4.29 Number System (1)

Practice Tests: Weekly Quiz 1 (15Q) Weekly Quiz 2 (15Q) Weekly Quiz 3 (15Q) Weekly Quiz 4 (15Q)

4.29.1 Number System: GATE CSE 1988 | Question: 2-vi

Define the value of r in the following: $\sqrt{(41)_r} = (7)_{10}$ 4

gate1988 digital-logic normal number-representation descriptive number-system 5

Answer key

4.30 Prime Implicants (2) 7

Practice Test: Test 1 (5Q) 8

4.30.1 Prime Implicants: GATE CSE 1997 | Question: 5.1

Let $f(x, y, z) = \bar{x} + \bar{y}x + xz$ be a switching function. Which one of the following is valid? 10

- A. $\bar{y}x$ is a prime implicant of f B. xz is a minterm of f 11
C. xz is an implicant of f D. y is a prime implicant of f

gate1997 digital-logic normal k-map prime-implicants 12

Answer key

4.30.2 Prime Implicants: GATE CSE 2004 | Question: 59

Which are the essential prime implicants of the following Boolean function? 13

$$f(a, b, c) = a'c + ac' + b'c$$
 14

- A. $a'c$ and ac' B. $a'c$ and $b'c$ C. $a'c$ only. D. ac' and $b'c$ 15

gatecse-2004 digital-logic normal prime-implicants

Answer key 16

4.31 ROM (4)

4.31.1 ROM: GATE CSE 1993 | Question: 6.6

A ROM is used to store the Truth table for binary multiple units that will multiply two 4-bit numbers. The size of the ROM (number of words $\times$ number of bits) that is required to accommodate the Truth table is M words $\times$ N bits. Write the values of M and N . 17

gate1993 digital-logic normal rom descriptive 19

Answer key

4.31.2 ROM: GATE CSE 1996 | Question: 1.21

A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is 20

- A. 256×16 B. $64K \times 8$ 22
C. $4K \times 16$ D. $64K \times 16$

gate1996 digital-logic normal rom

Answer key

4.31.3 ROM: GATE CSE 2012 | Question: 19



The amount of ROM needed to implement a 4-bit multiplier is ¹

- A. 64 bits B. 128 bits C. 1 Kbits D. 2 Kbits ²

gatecse-2012 digital-logic normal rom ³

Answer key

4.31.4 ROM: GATE IT 2004 | Question: 10



What is the minimum size of ROM required to store the complete truth table of an $8 - bit \times 8 - bit$ multiplier? ⁴

- A. $32K \times 16$ bits B. $64K \times 16$ bits ⁵
C. $16K \times 32$ bits D. $64K \times 32$ bits

gateit-2004 digital-logic normal rom ⁶

Answer key

4.32 Reduction (1)

4.32.1 Reduction: GATE CSE 2026 | Set 1 | Question: 11



Consider the following Boolean expression of a function F : ⁷

$$F(P, Q) = (\bar{P} + Q) \oplus (\bar{P}Q) \quad 8$$

Which of the following expressions is/are equivalent to F ? ⁹

- A. $\overline{P \oplus Q}$ B. $P \oplus Q$ ¹⁰
C. $\bar{P} \oplus Q$ D. $\bar{P} \oplus \bar{Q}$

gatecse-2026-set1 digital-logic boolean-algebra reduction multiple-selects one-mark ¹¹

Answer key

4.33 Ripple Counter Operation (1)

4.33.1 Ripple Counter Operation: GATE CSE 2025 | Set 2 | Question: 24



In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer) ¹²

gatecse2025-set2 digital-logic ripple-counter-operation numerical-answers one-mark ¹⁴

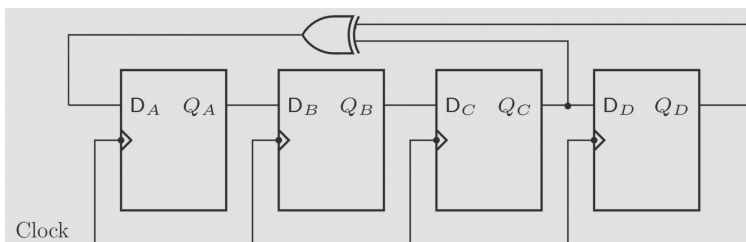
Answer key

4.34 Shift Registers (2)

4.34.1 Shift Registers: GATE CSE 1987 | Question: 13-a



The below figure shows four D-type flip-flops connected as a shift register using a XOR gate. The initial state and three subsequent states for three clock pulses are also given.



State	Q_A	Q_B	Q_C	Q_D
Initial	1	1	1	1
After the first clock	0	1	1	1
After the second clock	0	0	1	1
After the third clock	0	0	0	1

The state $Q_A Q_B Q_C Q_D$ after the fourth clock pulse is

- A. 0000 B. 1111 C. 1001 D. 1000

gate1987 digital-logic circuit-output sequential-circuit digital-counter shift-registers

Answer key

4.34.2 Shift Registers: GATE CSE 1991 | Question: 06,a

Using D flip-flop gates, design a parallel-in/serial-out shift register that shifts data from left to right with the following input lines:

- Clock CLK
- Three parallel data inputs A, B, C
- Serial input S
- Control input $\text{LOAD}/\overline{\text{SHIFT}}$.

gate1991 digital-logic difficult sequential-circuit flip-flop shift-registers descriptive

Answer key

4.35 Static Hazard (1)

4.35.1 Static Hazard: GATE CSE 2006 | Question: 38

Consider a Boolean function $f(w, x, y, z)$. Suppose that exactly one of its inputs is allowed to change at a time. If the function happens to be true for two input vectors $i_1 = \langle w_1, x_1, y_1, z_1 \rangle$ and $i_2 = \langle w_2, x_2, y_2, z_2 \rangle$, we would like the function to remain true as the input changes from i_1 to i_2 (i_1 and i_2 differ in exactly one bit position) without becoming false momentarily. Let $f(w, x, y, z) = \sum(5, 7, 11, 12, 13, 15)$. Which of the following cube covers of f will ensure that the required property is satisfied?

- $\bar{w}xz, wx\bar{y}, x\bar{y}z, xyz, wyz$
- $wxy, \bar{w}xz, wyz$
- $wx\bar{y}z, xz, w\bar{x}yz$
- $wx\bar{y}, wyz, wxz, \bar{w}xz, x\bar{y}z, xyz$

gatecse-2006 digital-logic min-sum-of-products-form difficult static-hazard

Answer key

4.36 Synchronous Asynchronous Circuits (4)

4.36.1 Synchronous Asynchronous Circuits: GATE CSE 1991 | Question: 03-ii

Advantage of synchronous sequential circuits over asynchronous ones is:

- faster operation
- ease of avoiding problems due to hazards
- lower hardware requirement
- better noise immunity
- none of the above

gate1991 digital-logic normal sequential-circuit synchronous-asynchronous-circuits multiple-selects

Answer key

4.36.2 Synchronous Asynchronous Circuits: GATE CSE 1998 | Question: 16



Design a synchronous counter to go through the following states: 1

1, 4, 2, 3, 1, 4, 2, 3, 1, 4, ...

gate1998 digital-logic normal descriptive synchronous-asynchronous-circuits 2

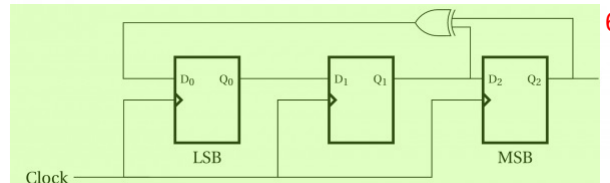
Answer key 3

4.36.3 Synchronous Asynchronous Circuits: GATE CSE 2001 | Question: 2.12

4



Consider the circuit given below with initial state $Q_0 = 1, Q_1 = Q_2 = 0$. The state of the circuit is given by 5 the value $4Q_2 + 2Q_1 + Q_0$



6

Which one of the following is correct state sequence of the circuit? 7

- A. 1, 3, 4, 6, 7, 5, 2 B. 1, 2, 5, 3, 7, 6, 4 8
C. 1, 2, 7, 3, 5, 6, 4 D. 1, 6, 5, 7, 2, 3, 4

gatecse-2001 digital-logic normal synchronous-asynchronous-circuits 9

Answer key 10

4.36.4 Synchronous Asynchronous Circuits: GATE CSE 2003 | Question: 44



11

A 1-input, 2-output synchronous sequential circuit behaves as follows: 12

Let z_k, n_k denote the number of 0's and 1's respectively in initial k bits of the input ($z_k + n_k = k$). The circuit outputs 00 until one of the following conditions holds.

- $z_k - n_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 10.
- $n_k - z_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 01.

What is the minimum number of states required in the state transition graph of the above circuit? 13

- A. 5 B. 6 C. 7 D. 8 14

gatecse-2003 digital-logic synchronous-asynchronous-circuits normal 15

Answer key

Answer Keys 16

4.1.1	N/A	17	4.1.2	N/A	4.1.3	B	4.1.4	B	4.1.5	A	18
4.1.6	B		4.1.7	19.2	4.1.8	B	4.1.9	-1	4.2.1	B	
4.2.2	C		4.3.1	C	4.4.1	D	4.4.2	C	4.4.3	N/A	
4.4.4	N/A		4.4.5	N/A	4.4.6	D	4.4.7	N/A	4.4.8	A	
4.4.9	B		4.4.10	D	4.4.11	C	4.4.12	D	4.4.13	C	
4.4.14	C		4.4.15	D	4.4.16	D	4.4.17	D	4.4.18	A	
4.4.19	A		4.4.20	D	4.4.21	D	4.4.22	1	4.4.23	A	
4.4.24	C		4.4.25	C	4.4.26	D	4.4.27	B	4.4.28	B;C;D	
4.4.29	B;C		4.4.30	B;D	4.4.31	B;C;D	4.4.32	B	4.4.33	B	
4.4.34	C		4.5.1	N/A	4.5.2	A	4.5.3	B	4.5.4	13:13	

4.5.5	A
4.6.3	C
4.6.8	3
4.8.1	B
4.8.6	A;C
4.8.11	D
4.8.16	B
4.8.21	A
4.8.26	C
4.8.31	D
4.8.36	X
4.9.1	C
4.11.3	B
4.12.5	A
4.13.3	N/A
4.13.8	C
4.13.13	A
4.13.18	N/A
4.15.4	A
4.17.3	A
4.18.1	C
4.18.6	C
4.19.3	N/A
4.20.1	D
4.20.6	-7.75 : -7.75
4.20.11	A;B;C
4.21.2	N/A
4.21.7	B
4.21.12	A
4.21.17	C
4.23.4	B
4.24.4	B
4.26.1	N/A
4.26.6	B
4.26.11	6 : 6
4.26.16	A
4.27.5	C
4.27.10	C
4.28.1	N/A
4.28.6	N/A

1

4.5.6	C
4.6.4	A
4.6.9	B
4.8.2	N/A
4.8.7	B
4.8.12	C
4.8.17	A
4.8.22	A
4.8.27	C
4.8.32	B
4.8.37	D
4.9.2	A;C
4.12.1	N/A
4.12.6	C
4.13.4	33.33
4.13.9	D
4.13.14	A
4.14.1	D
4.16.1	D
4.17.4	B
4.18.2	N/A
4.18.7	D
4.19.4	N/A
4.20.2	B
4.20.7	C
4.20.12	C
4.21.3	N/A
4.21.8	D
4.21.13	B
4.22.1	A;D
4.23.5	C
4.24.5	1
4.26.2	9
4.26.7	A
4.26.12	A;B
4.27.1	N/A
4.27.6	D
4.27.11	4
4.28.2	9
4.28.7	N/A

2

4.5.7	B
4.6.5	3
4.6.10	C;D
4.8.3	B
4.8.8	B
4.8.13	N/A
4.8.18	A
4.8.23	D
4.8.28	A
4.8.33	C
4.8.38	C
4.10.1	A
4.12.2	N/A
4.12.7	6:6
4.13.5	N/A
4.13.10	3
4.13.15	B
4.15.1	True
4.16.2	C
4.17.5	2
4.18.3	N/A
4.18.8	D
4.19.5	B;C
4.20.3	A
4.20.8	B
4.20.13	-60.25:-60.25
4.21.4	N/A
4.21.9	A
4.21.14	A;D
4.23.1	C
4.24.1	N/A
4.24.6	B
4.26.3	N/A
4.26.8	A
4.26.13	A
4.27.2	C
4.27.7	5
4.27.12	D
4.28.3	N/A
4.28.8	C

3

4.6.1	N/A
4.6.6	A
4.7.1	A
4.8.4	N/A
4.8.9	B
4.8.14	N/A
4.8.19	A
4.8.24	A
4.8.29	A
4.8.34	A;B;C
4.8.39	A
4.11.1	C
4.12.3	D
4.13.1	C
4.13.6	N/A
4.13.11	3:4
4.13.16	D
4.15.2	B
4.17.1	N/A
4.17.6	7:8
4.18.4	N/A
4.19.1	N/A
4.19.6	B
4.20.4	C
4.20.9	C
4.20.14	C
4.21.5	C
4.21.10	B
4.21.15	D
4.23.2	C
4.24.2	N/A
4.25.1	N/A
4.26.4	A
4.26.9	B
4.26.14	B;D
4.27.3	C
4.27.8	A
4.27.13	A
4.28.4	N/A
4.28.9	D

4

4.6.2	A
4.6.7	A
4.7.2	B
4.8.5	N/A
4.8.10	011
4.8.15	B
4.8.20	D
4.8.25	C
4.8.30	B
4.8.35	C
4.8.40	D
4.11.2	1034
4.12.4	D
4.13.2	N/A
4.13.7	A
4.13.12	B
4.13.17	D
4.15.3	B
4.17.2	C
4.17.7	D
4.18.5	N/A
4.19.2	B;C
4.19.7	A
4.20.5	B
4.20.10	B
4.21.1	D
4.21.6	C
4.21.11	C
4.21.16	A
4.23.3	B
4.24.3	B
4.25.2	A
4.26.5	B
4.26.10	3
4.26.15	B
4.27.4	A
4.27.9	B
4.27.14	N/A
4.28.5	N/A
4.28.10	C

2

4.28.11	D		4.28.12	C	4.28.13	A;C
4.28.16	C		4.28.17	B	4.28.18	D
4.28.21	N/A		4.28.22	A	4.28.23	D
4.28.26	A		4.28.27	A	4.28.28	B
4.28.31	A		4.28.32	B	4.28.33	5
4.28.36	-11		4.28.37	1	4.28.38	C
4.28.41	C		4.28.42	C	4.28.43	A
4.28.46	110		4.28.47	B	4.28.48	A;B;D
4.28.51	B;D		4.28.52	B	4.28.53	A
4.28.56	A		4.28.57	B	4.29.1	12
4.31.1	N/A		4.31.2	D	4.31.3	D
4.33.1	250:250		4.34.1	D	4.34.2	N/A
4.36.2	N/A		4.36.3	B	4.36.4	A

4.28.14	C		4.28.15	B
4.28.19	D		4.28.20	B
4.28.24	A		4.28.25	A
4.28.29	D		4.28.30	B
4.28.34	3		4.28.35	5
4.28.39	D		4.28.40	0.502 : 0.504
4.28.44	3 : 3		4.28.45	B
4.28.49	B;D		4.28.50	B;C
4.28.54	A		4.28.55	C
4.30.1	C		4.30.2	A
4.31.4	B		4.32.1	A;C
4.35.1	D		4.36.1	B



System calls, Processes, Threads, Inter-process communication, Concurrency and synchronization. 1
Deadlock. CPU scheduling. Memory management and Virtual memory. File systems. Disks is also under this

Mark Distribution in Previous GATE 2

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum 3
1 Mark Count	2	1	2	1	2	2	3	2	4	2	1
2 Marks Count	2	3	3	3	4	4	3	4	1	3	1
Total Marks	6	7	8	7	10	10	9	10	6	8	6

Welcome to the Operating System (OS) chapter of your GATE Computer Science preparation! This section is designed 4
as a comprehensive, exam-focused reference to help you master the core concepts, formulas, and problem-solving
techniques essential for excelling in the GATE exam. Operating Systems form the bedrock of computing, managing
hardware and software resources to provide a stable and efficient environment for applications. For GATE CS, OS is a
high-scoring subject, typically carrying a weightage of 8-12 marks. Questions range from conceptual understanding of
OS principles, numerical problems on scheduling, memory management, and disk I/O, to analytical questions on
deadlock and synchronization. A strong grasp of this subject is crucial not only for GATE but also for a fundamental
understanding of computer science.

Topic-wise Key Concepts 5

Bankers Algorithm 6

The Banker's Algorithm is a deadlock avoidance algorithm that checks for a safe state before granting a resource 7
request. It ensures that the system can always find a sequence of processes that can complete their execution without
leading to a deadlock.

• Important Formulas/Theorems: 8

1. **Need Matrix:** For each process P_i and resource type R_j , $Need[i][j] = Max[i][j] - Allocation[i][j]$. This 9
represents the remaining resources P_i may still request.

2. **Safety Algorithm:** 10

- Initialize $Work = Available$ and $Finish[i] = false$ for all processes. 11
- Find an i such that $Finish[i] == false$ and $Need[i] \leq Work$.
- If such an i exists, then $Work = Work + Allocation[i]$ and $Finish[i] = true$. Go to step 2.
- If no such i exists, the system is in a safe state if $Finish[i] == true$ for all i .

3. **Resource-Request Algorithm:** 12

- If $Request_i \leq Need_i$, proceed to step 2. Otherwise, error. 13
- If $Request_i \leq Available$, proceed to step 3. Otherwise, P_i must wait.
- Pretend to allocate resources: $Available = Available - Request_i$,
 $Allocation_i = Allocation_i + Request_i$, $Need_i = Need_i - Request_i$.
- Run Safety Algorithm. If safe, grant request. If unsafe, revert allocation and P_i waits.

• **Key Properties:** 14

- Ensures deadlock avoidance by maintaining the system in a safe state. 15
- Requires prior knowledge of maximum resource needs for each process.
- Less conservative than deadlock prevention, allowing higher resource utilization.

• **Common Pitfalls:** 16

- Confusing safe state with deadlock-free state (safe implies deadlock-free, but unsafe doesn't necessarily imply 17
deadlock).
- Incorrectly calculating the Need matrix or updating Available/Allocation vectors.
- Not checking all conditions in the Safety Algorithm or Resource-Request Algorithm.

• **Standard Problem-Solving Techniques:** 18

- Systematically trace the Safety Algorithm step-by-step, updating $Work$ and $Finish$ arrays. 19
- For resource requests, first check against $Need$ and then against $Available$, then temporarily allocate and run the Safety Algorithm.

Best Fit 20

Best Fit is a dynamic memory allocation algorithm that allocates the smallest available memory hole (partition) that is 21
large enough to satisfy a request. Its goal is to minimize the amount of internal fragmentation.

• Key Properties:1

- Tends to produce the smallest leftover hole, which might be useful for future small requests. 2
- Suffers from external fragmentation, where total free memory is sufficient but not contiguous.
- Requires searching the entire list of free partitions to find the best fit, making it slower than First Fit.

• Common Pitfalls:3

- Confusing it with First Fit or Worst Fit. 4
- Incorrectly identifying the "smallest" suitable hole, especially when multiple holes have the same size.

• Standard Problem-Solving Techniques:5

- Maintain a sorted list of free memory blocks (or iterate through all) and select the one with the smallest size greater than or equal to the request. 6
- Draw memory maps to visualize allocation and fragmentation.

Context Switch 7

A context switch is the mechanism by which the CPU saves the state of the currently executing process or thread and restores the state of another process or thread. This allows multiple processes/threads to share a single CPU. 8

• Key Properties:9

- Involves saving the Process Control Block (PCB) of the current process and loading the PCB of the next process. 10
- An overhead operation, as the CPU is not performing useful work during the switch.
- Latency is the time taken for a context switch.
- More frequent context switches (e.g., small time quantum in Round Robin) increase overhead.

• Common Pitfalls:

- Underestimating the overhead of context switching in performance calculations. 11
- Not understanding *what* exactly is saved (CPU registers, program counter, stack pointer, etc.).

DMA (Direct Memory Access) 12

DMA is a feature of computer systems that allows certain hardware subsystems (like disk drive controllers or network cards) to access main system memory (RAM) independently of the central processing unit (CPU). This significantly improves I/O performance by offloading data transfer from the CPU. 13

• Key Properties:14

- Reduces CPU overhead for large data transfers, allowing the CPU to perform other tasks. 15
- DMA controller manages the transfer, initiating an interrupt only upon completion.
- "Cycle stealing" occurs when the DMA controller temporarily takes control of the system bus from the CPU to transfer data.
- Requires careful setup by the CPU (source, destination, size).

• Common Pitfalls:16

- Thinking DMA eliminates CPU involvement entirely (CPU still initiates and handles completion). 17
- Not understanding the concept of cycle stealing and its impact on CPU performance.

Deadlock Prevention Avoidance Detection 18

These are strategies to deal with deadlocks. Prevention aims to ensure that at least one of the four necessary conditions for deadlock never holds. Avoidance dynamically grants resources only if the system remains in a safe state. Detection allows deadlocks to occur and then finds and resolves them. 19

• Important Formulas/Theorems: 20

1. Necessary Conditions for Deadlock (Coffman Conditions): 21

- C_1 : **Mutual Exclusion**: At least one resource must be held in a non-sharable mode. 22
- C_2 : **Hold and Wait**: A process holding at least one resource is waiting to acquire additional resources held by other processes.
- C_3 : **No Preemption**: Resources cannot be preempted; they can only be released voluntarily by the process holding them.
- C_4 : **Circular Wait**: A set of processes $\{P_0, P_1, \dots, P_n\}$ exists such that P_0 is waiting for a resource held by P_1 , P_1 is waiting for a resource held by P_2 , ..., P_{n-1} is waiting for a resource held by P_n , and P_n is waiting for a resource held by P_0 .

2. Deadlock Prevention: Break one or more of the four conditions. 23

- Break C_1 : Spooling (for printers), virtualizing resources. 24
- Break C_2 : Request all resources at once, or release all resources before requesting new ones.
- Break C_3 : Allow preemption of resources (e.g., CPU, memory).

- Break C_4 : Impose a total ordering of resource types; processes must request resources in increasing order. 1
- 3. **Deadlock Avoidance:** Banker's Algorithm (for multiple instances of resources), Resource-Allocation Graph Algorithm (for single instances). 2
- 4. **Deadlock Detection:** 3
 - If resources have single instances, use a Wait-For Graph. A cycle implies deadlock. 4
 - If resources have multiple instances, use an algorithm similar to Banker's Safety Algorithm.
- **Key Properties:** 5
 - Prevention is conservative, often leading to low resource utilization. 6
 - Avoidance is more flexible but requires future knowledge of resource requests.
 - Detection allows higher concurrency but incurs overhead for detection and recovery.
- **Common Pitfalls:** 7
 - Confusing which condition each prevention strategy addresses. 8
 - Incorrectly applying RAG for multiple instance resources (a cycle in RAG with multiple instances does not necessarily imply deadlock).
 - Not understanding the trade-offs between the three strategies.
- **Standard Problem-Solving Techniques:** 9
 - For prevention, identify which condition is being broken. 10
 - For avoidance, apply Banker's Algorithm.
 - For detection, draw the RAG or Wait-For Graph and look for cycles.

Demand Paging 11

Demand Paging is a virtual memory technique where pages are loaded into main memory only when they are referenced (demanded) by the CPU. This allows for larger virtual address spaces than physical memory and reduces I/O overhead by not loading unused pages. 12

- **Important Formulas/Theorems:** 13
 1. **Effective Access Time (EAT):**

$$EAT = (1 - p) \times MemoryAccessTime + p \times (PageFaultOverhead + MemoryAccessTime) \quad 14$$

Where p is the page fault rate. *PageFaultOverhead* includes time for interrupt, saving process state, reading page from disk, updating page table, restoring process state. 15

- **Key Properties:** 16
 - Leverages locality of reference to achieve good performance. 17
 - Reduces physical memory requirements for processes.
 - Introduces page faults, which are expensive operations.
 - Can lead to thrashing if the page fault rate becomes too high.
- **Common Pitfalls:** 18
 - Incorrectly calculating EAT, especially the components of page fault overhead. 19
 - Misunderstanding the concept of locality of reference.
 - Confusing demand paging with simple paging.
- **Standard Problem-Solving Techniques:** 20
 - Calculate EAT by plugging in given values for page fault rate, memory access time, and page fault overhead. 21
 - Analyze scenarios for thrashing.

Disk 22

A disk (hard disk drive) is a non-volatile storage device that stores data on rotating platters. It is characterized by its physical structure and the time it takes to access data. 23

- **Important Formulas/Theorems:** 24
 1. **Disk Access Time:** 25

$$AccessTime = SeekTime + RotationalLatency + TransferTime \quad 26$$

- **Seek Time:** Time taken for the disk arm to move the read/write heads to the correct track. 27
- **Rotational Latency:** Time taken for the desired sector to rotate under the read/write head (average is half a rotation).
- **Transfer Time:** Time taken to transfer the actual data.

$$TransferTime = \frac{NumberOfSectors}{SectorsPerTrack} \times RotationTime \quad 28$$

• Key Properties: 1

- Data is stored in sectors, which are grouped into tracks, and tracks are stacked into cylinders 2
- Seek time is typically the most significant component of disk access time.
- Mechanical components make disk access orders of magnitude slower than RAM access.

• Common Pitfalls: 3

- Incorrectly calculating average rotational latency (usually half a rotation). 4
- Forgetting to include all three components of access time.

• Standard Problem-Solving Techniques:

- Break down the problem into calculating each component (seek, rotational, transfer) and then sum them up. 5

Disk Scheduling 6

Disk scheduling algorithms aim to minimize the total head movement of the disk arm, thereby reducing the average disk access time and improving I/O throughput. They reorder pending disk I/O requests. 7

• Important Formulas/Theorems: 8

1. **Total Head Movement:** Sum of absolute differences between consecutive track numbers accessed. 9

• Algorithms:

- **FCFS (First-Come, First-Served):** Processes requests in the order they arrive. Simple but inefficient.
- **SSTF (Shortest Seek Time First):** Selects the request with the minimum seek time from the current head position. Can lead to starvation.
- **SCAN (Elevator Algorithm):** The disk arm moves from one end of the disk to the other, servicing requests along the way. When it reaches an end, it reverses direction.
- **C-SCAN (Circular SCAN):** Similar to SCAN, but when the arm reaches one end, it immediately returns to the other end without servicing requests on the return trip. Provides more uniform wait times.
- **LOOK:** Similar to SCAN, but the arm only goes as far as the last request in each direction, then reverses.
- **C-LOOK:** Similar to C-SCAN, but the arm only goes as far as the last request in each direction, then returns to the first request in the other direction without going to the absolute end.

• Key Properties:

- SSTF is optimal for average seek time but can cause starvation. 10
- SCAN/C-SCAN/LOOK/C-LOOK provide better fairness and prevent starvation.
- C-SCAN/C-LOOK are preferred for heavy loads as they provide more uniform service.

• Common Pitfalls: 11

- Incorrectly calculating head movement, especially for SCAN/C-SCAN/LOOK/C-LOOK where direction matters. 12
- Forgetting to include the initial head position in the calculation.
- Confusing SCAN with C-SCAN or LOOK with C-LOOK.

• Standard Problem-Solving Techniques: 13

- Draw a number line representing tracks and mark request positions. Trace the head movement for each algorithm. 14
- Carefully track the current head position and the direction of movement.

File System 15

A file system is an OS component that controls how data is stored and retrieved. It organizes files into a hierarchical structure, manages file attributes, and provides mechanisms for accessing and protecting files. 16

• Key Properties: 17

- **File Attributes:** Name, type, location, size, protection, time/date, user ID. 18
- **File Operations:** Create, write, read, reposition, delete, truncate.
- **Directory Structure:** Single-level, two-level, tree-structured, acyclic-graph, general graph.
- **Allocation Methods:** Contiguous, Linked, Indexed.
- **Free Space Management:** Bit vector, linked list, grouping, counting.

• Common Pitfalls: 19

- Confusing file allocation methods (e.g., linked vs. indexed). 20
- Not understanding the trade-offs of different directory structures (e.g., sharing, path length).

Fork System Call 21

The fork() system call creates a new process (child process) that is a duplicate of the calling process (parent process). 22 Both processes then execute concurrently from the point of the fork() call.

• Key Properties: 23

- The child process receives a copy of the parent's address space, including code, data, and stack segments (often implemented using copy-on-write). 1
- The child process gets a unique Process ID (PID).
- `fork()` returns 0 to the child process, the child's PID to the parent process, and -1 on failure.
- Child process inherits open file descriptors, signals, etc.

• Common Pitfalls: 2

- Incorrectly predicting the output of programs involving multiple `fork()` calls due to confusion about which process is executing which part of the code. 3
- Not understanding the return values of `fork()` in parent vs. child.
- Assuming parent and child share memory directly (they get copies, though copy-on-write optimizes this).

• Standard Problem-Solving Techniques: 4

- Trace the execution path for both parent and child processes separately after each `fork()`. 5
- Keep track of the number of processes created. For N `fork()` calls in a sequence, 2^N processes are created. If `fork()` is in a loop, it's more complex.

IO Handling 6

I/O handling refers to the mechanisms and techniques used by the operating system to manage data transfer between the CPU/memory and peripheral devices. Efficient I/O handling is crucial for system performance. 7

• Key Properties: 8

- **Polling:** CPU repeatedly checks the status of an I/O device. Simple but inefficient, wastes CPU cycles. 9
- **Interrupt-driven I/O:** Device notifies the CPU via an interrupt when I/O is complete or ready. More efficient for sporadic I/O.
- **DMA (Direct Memory Access):** Device controller transfers data directly to/from memory without CPU intervention. Most efficient for large data transfers.
- **Buffering:** Using temporary memory areas to hold data during transfer, smoothing out speed differences.
- **Spooling:** Holding data for a device (e.g., printer) in a buffer until the device is ready.

• Common Pitfalls: 10

- Confusing the efficiency and use cases of polling vs. interrupts vs. DMA.
- Not understanding the role of device controllers.

Input Output 11

Input/Output (I/O) refers to the communication between an information processing system (like a computer) and the outside world, possibly a human or another information processing system. It involves the transfer of data to and from peripheral devices. 12

• Key Properties: 13

- Can be synchronous (CPU waits for I/O completion) or asynchronous (CPU continues processing while I/O operates in background). 14
- Managed by device controllers and device drivers.
- Performance is often a bottleneck in overall system performance.

Inter Process Communication (IPC) 15

IPC refers to mechanisms provided by the operating system that allow independent processes to communicate and synchronize their actions. This is essential for cooperative processes and distributed systems. 16

• Key Properties: 17

- **Shared Memory:** Processes share a region of memory for direct data exchange. Fastest IPC, but requires explicit synchronization. 18
- **Message Passing:** Processes communicate by sending and receiving messages. Simpler to implement for small data transfers, easier to use in distributed systems.
- **Pipes:** Unidirectional (ordinary pipes) or bidirectional (named pipes/FIFOs) communication channels.
- **Sockets:** For network communication, allowing processes on different machines to communicate.
- **Semaphores/Mutexes/Monitors:** Primarily for synchronization, but can indirectly facilitate communication by coordinating access to shared resources.

• Common Pitfalls: 19

- Confusing shared memory with message passing (shared memory requires explicit synchronization, message passing often has built-in synchronization). 20
- Not understanding the difference between pipes and named pipes.

Interrupts 1

An interrupt is a signal to the CPU from a hardware device or a software program that indicates an event has occurred and requires immediate attention. It causes the CPU to temporarily suspend its current task and execute a special routine called an interrupt handler. 2

• Key Properties: 3

- **Hardware Interrupts:** Generated by I/O devices (e.g., disk completion, keyboard input). 4
- **Software Interrupts (Traps/Exceptions):** Generated by programs (e.g., system calls, division by zero, page fault).
- **Interrupt Vector:** A table containing the addresses of interrupt service routines (ISRs) for various interrupt types.
- **Maskable Interrupts:** Can be temporarily ignored by the CPU.
- **Non-maskable Interrupts (NMI):** Critical events that cannot be ignored (e.g., memory parity error).
- Involves saving the CPU's context (registers, PC) before executing the ISR and restoring it afterward.

• Common Pitfalls: 5

- Confusing hardware interrupts with software interrupts (traps). 6
- Not understanding the role of the interrupt vector.
- Forgetting the context saving/restoring overhead.

Least Recently Used (LRU) 7

LRU is a page replacement algorithm that replaces the page that has not been used for the longest period of time. It is based on the principle of temporal locality, assuming that pages recently used are likely to be used again soon. 8

• Key Properties: 9

- Generally performs well, approximating the optimal algorithm. 10
- Does not suffer from Belady's anomaly (increasing page faults with more frames).
- Difficult to implement efficiently in hardware, as it requires tracking the exact usage time of each page.
- Approximations like Second Chance or Clock algorithm are often used in practice.

• Common Pitfalls: 11

- Incorrectly identifying the "least recently used" page, especially in complex sequences. 12
- Confusing it with FIFO or LFU.

• Standard Problem-Solving Techniques: 13

- Maintain a "time stamp" or "counter" for each page, or conceptually keep a stack where the most recently used page is at the top. When a page is referenced, move it to the top. When a page fault occurs, remove the page at the bottom. 14
- Trace the page references and page frames step-by-step.

Linked Allocation 15

Linked allocation is a file allocation method where each file is stored as a linked list of disk blocks. Each block contains a pointer to the next block in the file. 16

• Key Properties: 17

- No external fragmentation, as any free block can be used. 18
- Files can grow dynamically.
- Sequential access is efficient, but direct access (random access) is very slow due to pointer traversal.
- Pointer overhead: a portion of each disk block is used for the pointer, reducing data storage capacity.
- Reliability issue: a lost or damaged pointer can lead to the loss of the rest of the file.

• Common Pitfalls: 19

- Confusing it with contiguous or indexed allocation. 20
- Overlooking the performance impact on random access or the pointer overhead.

Memory Management 21

Memory management is the operating system's function of handling primary memory. It allocates memory to processes, protects processes from each other, and provides an abstraction of a large, uniform memory space. 22

• Key Properties: 23

- **Relocation:** Mapping logical addresses to physical addresses. 24
- **Protection:** Preventing processes from accessing each other's memory.

- **Sharing:** Allowing processes to share memory regions.
- **Logical Organization:** Segmentation, paging.
- **Physical Organization:** Contiguous allocation, non-contiguous allocation.
- **Techniques:** Paging, segmentation, virtual memory.

• Common Pitfalls: 2

- Confusing internal vs. external fragmentation.
- Not understanding the difference between logical and physical addresses.

Multilevel Paging 4

Multilevel paging (or hierarchical paging) is a technique used in virtual memory systems to reduce the size of the page table. Instead of a single, large page table, it uses a tree-like structure of page tables, where the outer page table points to inner page tables, which then point to physical frames.

• Important Formulas/Theorems: 6

1. Effective Access Time (EAT) for N-level Paging: 7

$$EAT = (N + 1) \times MemoryAccessTime$$

(Assuming no TLB and page tables are in memory).

2. Number of Bits for Page Number: If a logical address is m bits and page size is 2^k bytes, then page offset is k bits, and page number is $(m - k)$ bits.

3. Number of Entries in Page Table: $2^{\text{page number bits}}$ 11

4. Size of Page Table: $2^{\text{page number bits}} \times \text{size of page table entry}$ 12

• Key Properties: 13

- Reduces the amount of physical memory required for page tables, especially for sparse address spaces.
- Increases memory access time because multiple memory accesses are needed to find a physical address (one for each level of page table).
- Can be combined with TLB to mitigate increased access time.

• Common Pitfalls: 15

- Incorrectly calculating the number of bits for each level of the page table.
- Forgetting to account for all memory accesses when calculating EAT without TLB.

• Standard Problem-Solving Techniques: 17

- Break down the virtual address into segments corresponding to each level of the page table and the page offset.
- Calculate the number of page table entries and the size of each page table level.

OS Protection 19

OS protection mechanisms ensure that system resources (CPU, memory, files, I/O devices) are used only in ways authorized by the operating system. This prevents malicious or erroneous programs from harming the system or other programs.

• Key Properties: 21

- **Dual-Mode Operation:** User mode (restricted privileges) and Kernel mode (full privileges). A mode bit indicates the current mode.
- **Privileged Instructions:** Instructions that can only be executed in kernel mode (e.g., I/O instructions, setting timer).
- **System Calls:** User programs request OS services, transitioning from user to kernel mode.
- **Memory Protection:** Base and limit registers, paging, segmentation.
- **Access Control Lists (ACLs):** For files and other resources, specifying who can access what and how.
- **Capabilities:** Tokens that grant specific access rights to a resource.

• Common Pitfalls: 23

- Confusing the purpose of dual-mode operation with memory protection.
- Not understanding how system calls facilitate a controlled transition to kernel mode.

Page Replacement 25

Page replacement algorithms decide which page to remove from main memory when a page fault occurs and no free frames are available. The goal is to minimize the number of page faults.

• Important Formulas/Theorems: 27

1. **Hit Ratio:** $\frac{\text{Number of Page Hits}}{\text{Total Page References}}$ 1
2. **Miss Ratio (Page Fault Rate):** $\frac{\text{Number of Page Faults}}{\text{Total Page References}}$
3. $\text{HitRatio} + \text{MissRatio} = 1$

• Algorithms: 2

- **FIFO (First-In, First-Out):** Replaces the page that has been in memory the longest. Simple but can suffer from Belady's anomaly. 3
- **Optimal (OPT/MIN):** Replaces the page that will not be used for the longest period of time in the future. Impossible to implement in practice, used as a benchmark.
- **LRU (Least Recently Used):** Replaces the page that has not been used for the longest time. Generally good performance, does not suffer from Belady's anomaly, but complex to implement.
- **LFU (Least Frequently Used):** Replaces the page with the smallest count of references.
- **MFU (Most Frequently Used):** Replaces the page with the largest count of references (less common).
- **Second Chance (Clock):** A FIFO variant that gives a page a "second chance" if its reference bit is set. 4

• Key Properties: 5

- Belady's anomaly: For some algorithms (like FIFO), increasing the number of available page frames can increase the number of page faults. 6
- Optimal algorithm provides the theoretical minimum number of page faults.

• Common Pitfalls: 7

- Incorrectly applying the rules for each algorithm, especially for LRU and Optimal. 8
- Forgetting to count the initial page loads as page faults.
- Misinterpreting Belady's anomaly.

• Standard Problem-Solving Techniques: 9

- Trace the page reference string through the page frames for each algorithm, step-by-step, marking hits and faults. 10
- Keep track of the state of each page (e.g., arrival time for FIFO, last used time for LRU, future use for Optimal).

Precedence Graph 11

A precedence graph (or process graph) is a directed acyclic graph (DAG) used to represent the execution order and dependencies among a set of tasks or processes. An edge from task A to task B means task A must complete before task B can start. 12

• Key Properties: 13

- Nodes represent tasks/processes, edges represent dependencies. 14
- Helps in identifying potential parallelism and critical paths.
- Used in scheduling to ensure correct execution order.
- Must be acyclic to represent a valid execution sequence.

• Common Pitfalls: 15

- Drawing cycles, which implies an impossible dependency. 16
- Misinterpreting the meaning of an edge (A → B means A *must* finish before B starts, not just that B follows A).

Process 17

A process is a program in execution. It is an active entity, unlike a program, which is a passive entity. Each process has its own address space, resources, and execution context. 18

• Key Properties: 19

- **Process Control Block (PCB):** Contains all information about a process (state, PC, registers, memory limits, open files, etc.). 20
- **Process States:**
 - **New:** Process being created.
 - **Ready:** Waiting to be assigned to a processor.
 - **Running:** Instructions are being executed.
 - **Waiting (Blocked):** Waiting for some event (e.g., I/O completion, signal).
 - **Terminated:** Process has finished execution.
- Each process has its own virtual address space, including code, data, and stack segments. 21
- Processes are typically heavy-weight compared to threads.

• Common Pitfalls: 22

- Confusing a process with a program or a thread. 23
- Incorrectly identifying transitions between process states.

Process Scheduling 1

Process scheduling is the activity of the operating system that selects which process to run next from the ready queue 2 and allocates the CPU to it. The goal is to optimize various performance metrics.

• Important Formulas/Theorems: 3

1. **Turnaround Time (TAT):** $CompletionTime - ArrivalTime$ 4
2. **Waiting Time (WT):** $TurnaroundTime - BurstTime$
3. **Response Time (RT):** $FirstResponseTime - ArrivalTime$
4. **Throughput:** Number of processes completed per unit time.
5. **CPU Utilization:** Percentage of time the CPU is busy.

• Algorithms: 5

- **FCFS (First-Come, First-Served):** Non-preemptive. Simple, but can have high average waiting time (convoy effect).
- **SJF (Shortest Job First):** Non-preemptive. Optimal for minimum average waiting time. Requires knowing future burst times.
- **SRTF (Shortest Remaining Time First):** Preemptive version of SJF. Optimal for minimum average waiting time.
- **Priority Scheduling:** Can be preemptive or non-preemptive. Processes with higher priority run first. Can lead to starvation (low-priority processes never run).
- **Round Robin (RR):** Preemptive. Each process gets a small unit of CPU time (time quantum). Fair, good for interactive systems.

• Key Properties: 7

- **Preemptive:** CPU can be taken away from a running process. 8
- **Non-preemptive:** CPU is held until process completes or blocks.
- **Starvation:** A process may wait indefinitely.
- **Aging:** Gradually increasing the priority of processes that wait for a long time to prevent starvation.

• Common Pitfalls: 9

- Incorrectly drawing Gantt charts, especially for preemptive algorithms. 10
- Mistakes in calculating TAT, WT, or RT.
- Confusing preemptive vs. non-preemptive versions of algorithms.
- Not handling arrival times correctly.

• Standard Problem-Solving Techniques: 11

- Draw a Gantt chart for each algorithm, carefully marking arrival, start, and completion times. 12
- Systematically calculate TAT, WT, and RT for each process and then their averages.
- For preemptive algorithms, check for preemption at every arrival or completion event.

Process Synchronization 13

Process synchronization refers to the coordination of multiple processes to ensure that they access shared resources or 14 critical sections in a controlled manner, preventing race conditions and maintaining data consistency.

• Important Formulas/Theorems: 15

1. Critical Section Problem Requirements: 16

- **Mutual Exclusion:** Only one process can be in its critical section at any time. 17
- **Progress:** If no process is in its critical section and some processes want to enter, only those not in their remainder section can participate in the decision, and this decision cannot be postponed indefinitely.
- **Bounded Waiting:** There is a limit on the number of times other processes can enter their critical sections after a process has made a request to enter its critical section and before that request is granted.

• Mechanisms: 18

- **Semaphores:** Integer variables accessed only through atomic wait() (P) and signal() (V) operations. Can be binary (mutex) or counting. 19
- **Mutex Locks:** A simpler binary semaphore, used for mutual exclusion.
- **Monitors:** High-level synchronization construct that encapsulates shared data and procedures that operate on the data, ensuring mutual exclusion and providing condition variables for waiting.
- **Hardware Solutions:** TestAndSet, Swap instructions. 20

• Key Properties: 21

- **Race Condition:** When multiple processes access and modify shared data concurrently, and the final result depends on the order of execution. 22
- **Deadlock:** Can occur with improper use of synchronization primitives (e.g., two processes waiting for each other's resources).

• Common Pitfalls: 23

- Incorrectly using wait() and signal() operations, leading to deadlock or starvation. 1
- Not ensuring atomicity of synchronization primitives.
- Confusing mutexes with counting semaphores.

• Standard Problem-Solving Techniques: 2

- Trace the execution flow of multiple processes accessing shared resources, observing the values of semaphores or mutexes. 3
- Identify potential race conditions and propose synchronization solutions.
- Analyze classic synchronization problems (Producer-Consumer, Readers-Writers, Dining Philosophers).

Resource Allocation 4

Resource allocation is the process by which the operating system assigns available resources (CPU, memory, I/O devices, files) to processes. This is a fundamental task of the OS, impacting system performance and stability. 5

• Key Properties: 6

- Resources can be preemptable (e.g., CPU) or non-preemptable (e.g., printer). 7
- Proper resource allocation is critical to prevent deadlocks and ensure fairness.
- Involves policies for granting requests and reclaiming resources.

Resource Allocation Graph (RAG) 8

A Resource Allocation Graph is a directed graph used to depict the state of a system regarding resources and processes. It helps in visualizing resource allocation and detecting deadlocks. 9

• Key Properties: 10

- **Nodes:** Processes (circles) and Resource types (rectangles). 11
- **Edges:**
 - **Request Edge:** From process to resource type (e.g., $P_i \rightarrow R_j$), indicating P_i is requesting an instance of R_j .
 - **Assignment Edge:** From resource type to process (e.g., $R_j \rightarrow P_i$), indicating an instance of R_j has been allocated to P_i .
- If the RAG contains a cycle, and each resource type has only one instance, then a deadlock exists. 12
- If the RAG contains a cycle, and some resource types have multiple instances, a deadlock *may* exist (further analysis needed). 13

• Common Pitfalls: 14

- Incorrectly drawing request vs. assignment edges. 15
- Assuming a cycle always implies deadlock, especially for multiple-instance resources.
- Not understanding the difference between RAG and Wait-For Graph.

• Standard Problem-Solving Techniques: 16

- Draw the RAG based on the given process and resource states. 17
- Identify cycles in the graph.
- If cycles exist, determine if they lead to deadlock based on resource instance counts.

Round Robin Scheduling 18

Round Robin (RR) is a preemptive process scheduling algorithm designed for time-sharing systems. Each process is given a small unit of CPU time, called a time quantum (or time slice). When the quantum expires, the process is preempted and added to the end of the ready queue. 19

• Important Formulas/Theorems: 20

1. **Turnaround Time (TAT):** $CompletionTime - ArrivalTime$ 21
2. **Waiting Time (WT):** $TurnaroundTime - BurstTime$
3. **Number of Context Switches:** Depends on the number of processes and the time quantum.

• Key Properties: 22

- Fair scheduling, as every process gets a chance to run. 23
- Good for interactive systems, providing quick response times.
- Performance heavily depends on the time quantum:
 - Small quantum: High context switch overhead, good response time.
 - Large quantum: Approaches FCFS, lower overhead, potentially poor response time.

• Common Pitfalls: 24

- Incorrectly handling the time quantum and preemption points. 25
- Mistakes in calculating TAT or WT, especially when processes arrive at different times.

- Forgetting to account for context switch overhead if specified in the problem. 1

• Standard Problem-Solving Techniques: 2

- Draw a Gantt chart, carefully allocating CPU time in quantum chunks and moving processes to the end of the ready queue upon preemption. 3
- Keep track of remaining burst times for each process.

Semaphore 4

A semaphore is a synchronization primitive, a protected integer variable that can only be accessed and modified by two atomic operations: wait() (also known as P or down) and signal() (also known as V or up). 5

• Important Formulas/Theorems: 6

1. wait(S) operation: 7

```
wait(S) {
    while (S <= 0); // busy wait
    S--;
```

(Alternatively, block the process if $S \leq 0$).

2. signal(S) operation: 8

```
signal(S) {
    S++;
}
```

(Alternatively, wake up a blocked process if any).

• Key Properties: 9

- **Counting Semaphore:** Can range over an unrestricted domain. Used to control access to a resource with multiple instances. 10
- **Binary Semaphore (Mutex):** Can only be 0 or 1. Used for mutual exclusion (like a lock).
- Can be used for both mutual exclusion and process synchronization (ordering of events).
- Improper use can lead to deadlock or starvation.

• Common Pitfalls: 11

- Swapping the order of wait() and signal() operations, leading to incorrect synchronization. 12
- Using busy-waiting semaphores in a single-processor system (wastes CPU cycles).
- Not initializing semaphores correctly.

• Standard Problem-Solving Techniques: 13

- Trace the values of semaphores as processes execute wait() and signal() operations. 14
- Identify critical sections and shared resources to determine appropriate semaphore usage.
- Solve classic synchronization problems using semaphores.

SRTF (Shortest Remaining Time First) 15

SRTF is a preemptive version of the Shortest Job First (SJF) scheduling algorithm. The CPU is allocated to the process with the smallest remaining burst time. If a new process arrives with a shorter burst time than the currently running process's remaining time, the current process is preempted. 16

• Important Formulas/Theorems: 17

1. **Turnaround Time (TAT):** $CompletionTime - ArrivalTime$ 18

2. **Waiting Time (WT):** $TurnaroundTime - BurstTime$

• Key Properties: 19

- Optimal for minimizing average waiting time. 20
- Requires knowing the remaining burst time of all processes, which is difficult in practice.
- High overhead due to frequent context switches, especially with many short jobs or frequent arrivals.
- Can suffer from starvation if a continuous stream of short jobs keeps arriving.

• Common Pitfalls: 21

- Incorrectly determining when preemption occurs (at every arrival or when a shorter job becomes ready). 22
- Mistakes in calculating remaining burst times.
- Overlooking processes that arrive later but have very short burst times.

• Standard Problem-Solving Techniques: 23

- Create a Gantt chart, carefully tracking the remaining burst time of all ready processes at each time unit. 24
- At each time unit, check for new arrivals and select the process with the minimum remaining burst time.

System Calls¹

System calls provide an interface between a running program and the operating system. They allow user-level programs to request services from the OS kernel, such as file I/O, process creation, memory allocation, and device access.²

• Key Properties:³

- Transition from user mode to kernel mode (privileged mode).⁴
- Typically invoked through a software interrupt (trap).
- Provide a controlled and secure way for user programs to access protected resources.
- Examples: `fork()`, `exec()`, `read()`, `write()`, `open()`, `close()`, `wait()`, `exit()`.

• Common Pitfalls:⁵

- Confusing system calls with library calls (library calls are user-level functions that may or may not invoke system calls).⁶
- Not understanding the mode switch involved.

Threads⁷

A thread (or lightweight process) is a basic unit of CPU utilization within a process. Multiple threads within the same process share the same code section, data section, and other OS resources (like open files, signals), but each has its own program counter, register set, and stack.⁸

• Key Properties:⁹

- **Faster Context Switching:** Switching between threads in the same process is faster than switching between processes.¹⁰
- **Resource Sharing:** Threads within a process share memory and resources, facilitating communication.
- **Concurrency:** Multiple threads can execute concurrently, improving application responsiveness and throughput on multi-core systems.
- **User-level Threads:** Managed by a user-level library, faster to create/manage, but if one thread blocks, the entire process blocks.
- **Kernel-level Threads:** Managed by the OS kernel, slower to create/manage, but if one thread blocks, others can still run.
- **Many-to-One, One-to-One, Many-to-Many models:** Different mappings between user and kernel threads.

• Common Pitfalls:¹¹

- Confusing threads with processes (threads share memory, processes do not).¹²
- Forgetting that synchronization is still required for shared data among threads.
- Misunderstanding the implications of user-level vs. kernel-level threads.

Translation Lookaside Buffer (TLB)¹³

The TLB is a small, fast hardware cache that stores recent virtual-to-physical address translations. It is used to speed up memory access in paged virtual memory systems by avoiding multiple memory accesses to the page table.¹⁴

• Important Formulas/Theorems:¹⁵

1. Effective Access Time (EAT) with TLB:¹⁶

$$EAT = TLB_Hit_Ratio \times (TLB_Access_Time + Memory_Access_Time) + (1 - TLB_Hit_Ratio) \times Page_Table_Access_Time$$
¹⁷

Where *Page_Table_Access_Time* is the time to access the page table in main memory. If the page table is also in memory, this is another *Memory_Access_Time*.¹⁸

• Key Properties:¹⁹

- Significantly reduces the average memory access time if the hit ratio is high.²⁰
- A TLB miss requires a page table walk (accessing the page table in main memory).
- TLB entries are flushed on context switches between processes (though some TLBs support ASIDs to avoid this).

• Common Pitfalls:²¹

- Incorrectly calculating EAT, especially the components for a TLB miss (TLB access + memory access for page table + memory access for data).²²
- Forgetting to include TLB access time even on a hit.

• Standard Problem-Solving Techniques:²³

- Carefully plug values into the EAT formula, ensuring all access times are accounted for.²⁴
- Understand how a TLB miss impacts the total access time.

Virtual Memory¹

Virtual memory is a memory management technique that provides an application with an illusion of a contiguous, large address space, even if the physical memory is fragmented or smaller than the virtual address space. It separates logical memory from physical memory.²

• Key Properties:³

- Allows programs to be larger than physical memory.⁴
- Enables efficient sharing of memory among multiple processes.
- Provides memory protection.
- Commonly implemented using paging (demand paging) and swapping.
- Introduces the concept of page faults.

• Important Formulas/Theorems:⁵

1. **Effective Access Time (EAT):** (See Demand Paging and TLB sections for detailed formulas).⁶

• Common Pitfalls:⁷

- Confusing virtual memory with physical memory.
- Not understanding the role of the MMU (Memory Management Unit) in address translation.
- Misinterpreting the benefits and overheads of virtual memory.

Quick Formula Reference⁸

• Banker's Algorithm:⁹

- Need Matrix: $Need[i][j] = Max[i][j] - Allocation[i][j]$

• Demand Paging / Virtual Memory EAT (without TLB):¹⁰

$$EAT = (1 - p) \times MemoryAccessTime + p \times (PageFaultOverhead + MemoryAccessTime) \quad 11$$

where p is page fault rate.¹²

• Disk Access Time:¹³

$$AccessTime = SeekTime + RotationalLatency + TransferTime \quad 14$$

$$AverageRotationalLatency = \frac{1}{2} \times RotationTime \quad 15$$

$$TransferTime = \frac{NumberOfSectors}{SectorsPerTrack} \times RotationTime \quad 16$$

• Multilevel Paging EAT (without TLB):¹⁷

$$EAT = (N + 1) \times MemoryAccessTime \quad 18$$

for N -level paging, assuming page tables are in memory.¹⁹

• Page Replacement:²⁰

- Hit Ratio: $\frac{\text{Number of Page Hits}}{\text{Total Page References}}$
- Miss Ratio (Page Fault Rate): $\frac{\text{Number of Page Faults}}{\text{Total Page References}}$
- $HitRatio + MissRatio = 1$

• Process Scheduling Metrics:²¹

- Turnaround Time (TAT): $CompletionTime - ArrivalTime$
- Waiting Time (WT): $TurnaroundTime - BurstTime$
- Response Time (RT): $FirstResponseTime - ArrivalTime$

• TLB EAT:²²

$$EAT = TLB_Hit_Ratio \times (TLB_Access_Time + Memory_Access_Time) + (1 - TLB_Hit_Ratio) \times 23$$

Important Tips for GATE²⁴

1. **Master Numerical Problems:** A significant portion of OS questions in GATE are numerical, especially from Process Scheduling (FCFS, SJF, SRTF, RR), Memory Management (Paging, Page Replacement, TLB EAT), and Disk Scheduling. Practice drawing Gantt charts and calculating metrics meticulously.²⁵
2. **Understand Preemption:** Pay close attention to whether a scheduling algorithm is preemptive or non-preemptive.

This is a common source of error in Gantt chart construction. For preemptive algorithms, always check for new arrivals at each time unit. 1

3. **Trace Algorithms Systematically:** For complex algorithms like Banker's, Page Replacement, or Disk Scheduling, trace the steps methodically. Draw diagrams (RAG, memory maps, Gantt charts) to visualize the state changes. 2

4. **Differentiate Similar Concepts:** Be clear on the distinctions between processes and threads, mutexes and semaphores, internal and external fragmentation, logical and physical addresses, and different deadlock strategies (prevention, avoidance, detection). 3

5. **Read Questions Carefully:** Look for keywords like "average," "minimum," "maximum," "at least," "at most," "preemptive," "non-preemptive," and specific units (ms, ns). A small detail can change the entire answer. 4

6. **Focus on Trade-offs:** Many OS concepts involve trade-offs (e.g., performance vs. fairness in scheduling, memory usage vs. access time in paging). Understand why certain design choices are made and their implications. 5

7. **Review Formulas Regularly:** Keep the key formulas for EAT, scheduling metrics, and disk access time handy and review them frequently. Understand the components of each formula. 6

8. **Practice with GATE Previous Year Questions:** This will help you understand the common question patterns, difficulty levels, and time management strategies specific to the GATE exam. 7

5.1

Bankers Algorithm (2)

 **Practice Test:** Test 1 (8Q) 9

5.1.1 Bankers Algorithm: GATE CSE 2018 | Question: 39



In a system, there are three types of resources: E , F and G . Four processes P_0 , P_1 , P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $\text{Max}[P_2, F]$ is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation. 10

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are only resources available. 11

Allocation				Max			
	E	F	G		E	F	G
P_0	1	0	1	P_0	4	3	1
P_1	1	1	2	P_1	2	1	4
P_2	1	0	3	P_2	1	3	3
P_3	2	0	0	P_3	5	4	1

 12

From the perspective of deadlock avoidance, which one of the following is true? 13

- A. The system is in *safe* state
 - B. The system is not in *safe* state, but would be *safe* if one more instance of E were available
 - C. The system is not in *safe* state, but would be *safe* if one more instance of F were available
 - D. The system is not in *safe* state, but would be *safe* if one more instance of G were available
- 14

gatecse-2018 operating-system deadlock-prevention-avoidance-detection bankers-algorithm normal two-marks

Answer key 

5.1.2 Bankers Algorithm: GATE CSE 2026 | Set 1 | Question: 19



With respect to deadlocks in an operating system, which of the following statements is/are FALSE?

- A. Banker's algorithm is used to prevent deadlocks
- B. Deadlock formation can be prevented by ensuring that the hold and wait condition is not allowed
- C. An assignment edge in a resource allocation graph is marked from a process to a resource
- D. A safe state guarantees that all processes can finish without formation of a deadlock

gatecse-2026-set1 operating-system deadlock-prevention-avoidance-detection bankers-algorithm multiple-selects one-mark

Answer key 

5.2.1 Best Fit: GATE CSE 2026 | Set 2 | Question: 45



Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (answer in integer)

Note: $1K = 2^{10}$

gatecse-2026-set2 operating-system memory-management best-fit numerical-answers two-marks 2

Answer key

Practice Test: Test 1 (6Q) 3

5.3.1 Context Switch: GATE CSE 1999 | Question: 2.12



Which of the following actions is/are typically not performed by the operating system when switching context from process *A* to process *B*?

- A. Saving current register values and restoring saved register values for process *B*.
- B. Changing address translation tables.
- C. Swapping out the memory image of process *A* to the disk.
- D. Invalidating the translation look-aside buffer.

gate1999 operating-system context-switch normal

Answer key

5.3.2 Context Switch: GATE CSE 2000 | Question: 1.20, ISRO2008-47



Which of the following need not necessarily be saved on a context switch between processes?

- A. General purpose registers
- B. Translation look-aside buffer
- C. Program counter
- D. All of the above

gatecse-2000 operating-system easy isro2008 context-switch 8

Answer key

5.3.3 Context Switch: GATE CSE 2011 | Question: 6, UGCNET-June2013-III: 62



Let the time taken to switch from user mode to kernel mode of execution be T_1 while time taken to switch between two user processes be T_2 . Which of the following is correct?

- A. $T_1 > T_2$
- B. $T_1 = T_2$
- C. $T_1 < T_2$
- D. Nothing can be said about the relation between T_1 and T_2

gatecse-2011 operating-system context-switch easy ugcnetcse-june2013-paper3 12

Answer key

5.3.4 Context Switch: GATE CSE 2024 | Set 2 | Question: 15



Consider a process *P* running on a CPU. Which one or more of the following events will *a/ways* trigger a context switch by the OS that results in process *P* moving to a non-running state (e.g., ready, blocked)?

- A. *P* makes a blocking system call to read a block of data from the disk
- B. *P* tries to access a page that is in the swap space, triggering a page fault
- C. An interrupt is raised by the disk to deliver data requested by some other process
- D. A timer interrupt is raised by the hardware

Answer key

5.4

DMA (1)

5.4.1 DMA: GATE CSE 2001 | Question: 8



Consider a disk with the following specifications: 20 surfaces, 1000 tracks/surface, 16 sectors/track, data density 1 KB/sector, rotation speed 3000 rpm. The operating system initiates the transfer between the disk and the memory sector-wise. Once the head has been placed on the right track, the disk reads a sector in a single scan. It reads bits from the sector while the head is passing over the sector. The read bits are formed into bytes in a serial-in-parallel-out buffer and each byte is then transferred to memory. The disk writing is exactly a complementary process.

For parts (C) and (D) below, assume memory read-write time = 0.1 microseconds/byte, interrupt driven transfer has an interrupt overhead = 0.4 microseconds, the DMA initialization, and termination overhead is negligible compared to the total sector transfer time. DMA requests are always granted.

- What is the total capacity of the disk?
- What is the data transfer rate?
- What is the percentage of time the CPU is required for this disk I/O for byte-wise interrupts driven transfer?
- What is the maximum percentage of time the CPU is held up for this disk I/O for cycle-stealing DMA transfer?

Answer key

5.5

Deadlock Prevention Avoidance Detection (4)

Practice Test: Test 1 (9Q)

5.5.1 Deadlock Prevention Avoidance Detection: GATE CSE 2018 | Question: 24



Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resources can be requested and releases only one at a time. The largest value of K that will always avoid deadlock is ____

Answer key

5.5.2 Deadlock Prevention Avoidance Detection: GATE CSE 2021 | Set 2 | Question: 43



Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance:

```
function OWNRESOURCE(Resource R)
  Acquire lock L // a global lock
  if R is available then
    Acquire R
    Release lock L
  else
    if R is owned by another process P then
      Terminate P, after releasing all resources owned by P
    Acquire R
    Restart P
    Release lock L
  end if
end if
end function
```

Which of the following choice(s) about the above scheme is/are correct? 8

- The scheme ensures that deadlocks will not occur 9
- The scheme may lead to live-lock
- The scheme may lead to starvation
- The scheme violates the mutual exclusion property

Answer key

5.5.3 Deadlock Prevention Avoidance Detection: GATE CSE 2026 | Set 1 | Question: 25



Consider a system consisting of k instances of a resource R , being shared by 5 processes. Assume that each process requires a maximum of two instances of resource R and a process can request or release only one instance at a time. Further, a process can request the second instance of the resource only after acquiring the first instance.

The minimum value of k for the system to be deadlock-free is _____. (answer in integer)

Answer key

5.5.4 Deadlock Prevention Avoidance Detection: GATE IT 2004 | Question: 63

4



In a certain operating system, deadlock prevention is attempted using the following scheme. Each process is assigned a unique timestamp, and is restarted with the same timestamp if killed. Let P_h be the process holding a resource R , P_r be a process requesting for the same resource R , and $T(P_h)$ and $T(P_r)$ be their timestamps respectively. The decision to wait or preempt one of the processes is based on the following algorithm.

```
if  $T(P_r) < T(P_h)$  then
    kill  $P_r$ 
else wait
```

Which one of the following is TRUE?

- A. The scheme is deadlock-free, but not starvation-free
- B. The scheme is not deadlock-free, but starvation-free
- C. The scheme is neither deadlock-free nor starvation-free
- D. The scheme is both deadlock-free and starvation-free

Answer key

5.6 Demand Paging (3)

5.6.1 Demand Paging: GATE CSE 2022 | Question: 54



Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string

7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1

the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____.

Answer key

5.6.2 Demand Paging: GATE CSE 2025 | Set 1 | Question: 4



Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?

- A. 2^{21}
- B. 2^{20}
- C. 2^{22}
- D. 2^{24}

Answer key

5.6.3 Demand Paging: GATE CSE 2025 | Set 2 | Question: 37



Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 6 4 5 1 3 2 . The contents of the frames are as follows initially and after each reference (from left to right):

initially	after												
—	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
—	1	1	1	1	1	1	1	6	6	6	6	6	2
—	—	2	2	4	4	4	4	4	4	4	1	1	1
—	—	—	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

- A. Least Recently Used page replacement policy
- B. Least Frequently Used page replacement policy
- C. Most Frequently Used page replacement policy
- D. Optimal page replacement policy

gatecse2025-set2 operating-system page-replacement demand-paging multiple-selects two-marks

Answer key

5.7 Disk (30)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (2Q)

5.7.1 Disk: GATE CSE 1990 | Question: 7-c



A certain moving arm disk-storage device has the following specifications:

- Number of tracks per surface = 404
- Track storage capacity = 130030 bytes.
- Disk speed = 3600 rpm
- Average seek time = 30 m secs.

Estimate the average latency, the disk storage capacity, and the data transfer rate.

gate1990 operating-system disk descriptive

Answer key

5.7.2 Disk: GATE CSE 1993 | Question: 6.7



A certain moving arm disk storage, with one head, has the following specifications:

- Number of tracks/recording surface = 200
- Disk rotation speed = 2400 rpm
- Track storage capacity = 62,500 bits

The average latency of this device is P ms and the data transfer rate is Q bits/sec. Write the values of P and Q.

gate1993 operating-system disk normal descriptive

Answer key

5.7.3 Disk: GATE CSE 1993 | Question: 7.8



The root directory of a disk should be placed

- A. at a fixed address in main memory
- B. at a fixed location on the disk
- C. anywhere on the disk
- D. at a fixed location on the system disk

E. anywhere on the system disk **1**

gate1993 operating-system disk normal **2**

Answer key

5.7.4 Disk: GATE CSE 1995 | Question: 14



If the overhead for formatting a disk is 96 bytes for a 4000 byte sector, **3**

A. Compute the unformatted capacity of the disk for the following parameters: **4**

- Number of surfaces: 8
- Outer diameter of the disk: 12 cm
- Inner diameter of the disk: 4 cm
- Inner track space: 0.1 mm
- Number of sectors per track: 20

B. If the disk in (A) is rotating at 360 rpm, determine the effective data transfer rate which is defined as the number **5** of bytes transferred per second between disk and memory.

gate1995 operating-system disk normal descriptive

Answer key

5.7.5 Disk: GATE CSE 1996 | Question: 23



A file system with a one-level directory structure is implemented on a disk with disk block size of $4K$ bytes. **6**
The disk is used as follows:

Disk-block 0	File Allocation Table, consisting of one 8-bit entry per data block, representing the data block address of the next data block in the file	7
Disk-block 1	Directory, with one 32 bit entry per file:	
Disk-block 2	Data-block 1;	
Disk-block 3	Data-block 2; etc.	

- a. What is the maximum possible number of files? **8**
b. What is the maximum possible file size in blocks

gate1996 operating-system disk normal file-system descriptive

Answer key

5.7.6 Disk: GATE CSE 1997 | Question: 74



A program P reads and processes 1000 consecutive records from a sequential file F stored on device D **9** without using any file system facilities. Given the following

- Size of each record = 3200 bytes **10**
- Access time of $D = 10$ msec
- Data transfer rate of $D = 800 \times 10^3$ bytes/second
- CPU time to process each record = 3 msec

What is the elapsed time of P if **11**

- A. F contains unblocked records and P does not use buffering? **12**
B. F contains unblocked records and P uses one buffer (i.e., it always reads ahead into the buffer)?
C. records of F are organized using a blocking factor of 2 (i.e., each block on D contains two records of F) and P uses one buffer?

gate1997 operating-system disk

Answer key

5.7.7 Disk: GATE CSE 1998 | Question: 2-9



Formatting for a floppy disk refers to **1**

- A. arranging the data on the disk in contiguous fashion **2**
- B. writing the directory
- C. erasing the system data
- D. writing identification information on all tracks and sectors

gate1998 operating-system disk normal **3**

Answer key

5.7.8 Disk: GATE CSE 1998 | Question: 25-a



Free disk space can be used to keep track of using a free list or a bit map. Disk addresses require d bits. For a disk with B blocks, F of which are free, state the condition under which the free list uses less space than the bit map.

gate1998 operating-system disk descriptive **6**

Answer key

5.7.9 Disk: GATE CSE 1998 | Question: 25b



Consider a disk with c cylinders, t tracks per cylinder, s sectors per track and a sector length s_l . A logical file d_l with fixed record length r_l is stored continuously on this disk starting at location (c_L, t_L, s_L) , where c_L, t_L and s_L are the cylinder, track and sector numbers, respectively. Derive the formula to calculate the disk address (i.e. cylinder, track and sector) of a logical record n assuming that $r_l = s_l$. **8**

gate1998 operating-system disk descriptive **9**

Answer key

5.7.10 Disk: GATE CSE 1999 | Question: 2-18, ISRO2008-46



Raid configurations of the disks are used to provide **10**

- A. Fault-tolerance
- B. High speed **11**
- C. High data density
- D. (A) & (B)

gate1999 operating-system disk easy isro2008 **12**

Answer key **13**

5.7.11 Disk: GATE CSE 2001 | Question: 1.22



Which of the following requires a device driver? **15**

- A. Register
- B. Cache
- C. Main memory
- D. Disk **16**

gatecse-2001 operating-system disk easy **17**

Answer key **18**

5.7.12 Disk: GATE CSE 2003 | Question: 25, ISRO2009-12



Using a larger block size in a fixed block size file system leads to

- A. better disk throughput but poorer disk space utilization
- B. better disk throughput and better disk space utilization
- C. poorer disk throughput but better disk space utilization
- D. poorer disk throughput and poorer disk space utilization

gatecse-2003 operating-system disk normal isro2009

Answer key

5.7.13 Disk: GATE CSE 2004 | Question: 49



A unix-style I-nodes has 10 direct pointers and one single, one double and one triple indirect pointers. Disk block size is 1 Kbyte, disk block address is 32 bits, and 48-bit integers are used. What is the maximum possible file size?

- A. 2^{24} bytes B. 2^{32} bytes C. 2^{34} bytes D. 2^{48} bytes 2

gatecse-2004 operating-system disk normal 3

Answer key 4

5.7.14 Disk: GATE CSE 2005 | Question: 21



What is the swap space in the disk used for? 5

- A. Saving temporary html pages B. Saving process data 6
C. Storing the super-block D. Storing device drivers

gatecse-2005 operating-system disk easy 7

Answer key

5.7.15 Disk: GATE CSE 2007 | Question: 11, ISRO2009-36, ISRO2016-21



Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

- A. 256 Mbyte, 19 bits B. 256 Mbyte, 28 bits 10
C. 512 Mbyte, 20 bits D. 64 Gbyte, 28 bits

gatecse-2007 operating-system disk normal isro2016 11

Answer key 12

5.7.16 Disk: GATE CSE 2008 | Question: 32



For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to 13

- A. non-uniform distribution of requests 15
B. arm starting and stopping inertia
C. higher capacity of tracks on the periphery of the platter
D. use of unfair arm scheduling policies

gatecse-2008 operating-system disk normal 16

Answer key

5.7.17 Disk: GATE CSE 2009 | Question: 51



A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple $\langle c, h, s \rangle$, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0^{th} sector is addresses as $\langle 0, 0, 0 \rangle$, the 1^{st} sector as $\langle 0, 0, 1 \rangle$, and so on. The address $\langle 400, 16, 29 \rangle$ corresponds to sector number:

- A. 505035 B. 505036 C. 505037 D. 505038 18

gatecse-2009 operating-system disk normal 19

Answer key 20

5.7.18 Disk: GATE CSE 2009 | Question: 52



A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple $\langle c, h, s \rangle$, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0^{th} sector is addresses as $\langle 0, 0, 0 \rangle$, the 1^{st} sector as $\langle 0, 0, 1 \rangle$, and so on

The address of the 1039th sector is 1

- A. $\langle 0, 15, 31 \rangle$ B. $\langle 0, 16, 30 \rangle$ 2
C. $\langle 0, 16, 31 \rangle$ D. $\langle 0, 17, 31 \rangle$

gatecse-2009 operating-system disk normal 3

Answer key

5.7.19 Disk: GATE CSE 2011 | Question: 44



An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected.) 4

- A. 0.50 s B. 1.50 s C. 1.25 s D. 1.00 s 5

gatecse-2011 operating-system disk normal 6

Answer key

5.7.20 Disk: GATE CSE 2012 | Question: 41



A file system with 300 GByte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

- A. 3 KBytes B. 35 KBytes 9
C. 280 KBytes D. dependent on the size of the disk

gatecse-2012 operating-system disk normal 10

Answer key

5.7.21 Disk: GATE CSE 2013 | Question: 29



Consider a hard disk with 16 recording surfaces (0 – 15) having 16384 cylinders (0 – 16383) and each cylinder contains 64 sectors (0 – 63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is $\langle \text{cylinder no.}, \text{surface no.}, \text{sector no.} \rangle$. A file of size 42797 KB is stored in the disk and the starting disk location of the file is $\langle 1200, 9, 40 \rangle$. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner? 12

- A. 1281 B. 1282 C. 1283 D. 1284 13

gatecse-2013 operating-system disk normal 14

Answer key

5.7.22 Disk: GATE CSE 2014 | Set 2 | Question: 20



A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____. 15

gatecse-2014-set2 operating-system disk numerical-answers normal file-system 16

Answer key

5.7.23 Disk: GATE CSE 2015 | Set 1 | Question: 48



Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is _____

gatecse-2015-set1 operating-system disk normal numerical-answers

Answer key

5.7.24 Disk: GATE CSE 2015 | Set 2 | Question: 49



Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is _____ 1

gatecse-2015-set2 operating-system disk normal numerical-answers

Answer key

5.7.25 Disk: GATE CSE 2024 | Set 1 | Question: 44



Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____. 2

gatecse-2024-set1 numerical-answers operating-system disk two-marks 3

Answer key

5.7.26 Disk: GATE CSE 2024 | Set 2 | Question: 43



Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____. 4

gatecse-2024-set2 numerical-answers operating-system disk two-marks 5

Answer key

5.7.27 Disk: GATE IT 2005 | Question: 63



In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both. What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively? 6

- A. 35400 and 35800 bytes
B. 35800 and 35400 bytes 8
C. 35600 and 35400 bytes
D. 35400 and 35600 bytes

gateit-2005 operating-system disk normal 9

Answer key

5.7.28 Disk: GATE IT 2005 | Question: 81-a



A disk has 8 equidistant tracks. The diameters of the innermost and outermost tracks are 1 cm and 8 cm respectively. The innermost track has a storage capacity of 10 MB. 10

What is the total amount of data that can be stored on the disk if it is used with a drive that rotates it with 11

- I. Constant Linear Velocity 12
II. Constant Angular Velocity?

- A. I. 80 MB; II. 2040 MB
B. I. 2040 MB; II. 80 MB 13
C. I. 80 MB; II. 360 MB
D. I. 360 MB; II. 80 MB

gateit-2005 operating-system disk normal 14

Answer key

5.7.29 Disk: GATE IT 2005 | Question: 81-b



A disk has 8 equidistant tracks. The diameters of the innermost and outermost tracks are 1 cm and 8 cm respectively. The innermost track has a storage capacity of 10 MB.

If the disk has 20 sectors per track and is currently at the end of the 5th sector of the inner-most track and the head can move at a speed of 10 meters/sec and it is rotating at constant angular velocity of 6000 RPM, how much time will it take to read 1 MB contiguous data starting from the sector 4 of the outer-most track? 1

- A. 13.5 ms B. 10 ms C. 9.5 ms D. 20 ms 2

gateit-2005 operating-system disk normal 3

Answer key

5.7.30 Disk: GATE IT 2007 | Question: 44, ISRO2015-34



A hard disk system has the following parameters : 4

- Number of tracks = 500
 - Number of sectors/track = 100
 - Number of bytes /sector = 500
 - Time taken by the head to move from one track to adjacent track = 1 ms
 - Rotation speed = 600 rpm.
- 5

What is the average time taken for transferring 250 bytes from the disk ? 6

- A. 300.5 ms B. 255.5 ms C. 255 ms D. 300 ms 7

gateit-2007 operating-system disk normal isro2015 8

Answer key 9

5.8 10

Disk Scheduling (16) 11

Practice Test: Test 1 (14Q)

5.8.1 Disk Scheduling: GATE CSE 1989 | Question: 4-xii



Disk requests come to disk driver for cylinders 10, 22, 20, 2, 40, 6 and 38, in that order at a time when the disk drive is reading from cylinder 20. The seek time is 6 msec per cylinder. Compute the total seek time if the disk arm scheduling algorithm is. 12

- A. First come first served. 13
B. Closest cylinder next.

gate1989 descriptive operating-system disk-scheduling 14

Answer key

5.8.2 Disk Scheduling: GATE CSE 1990 | Question: 9b



Assuming the current disk cylinder to be 50 and the sequence for the cylinders to be 1, 36, 49, 65, 53, 12, 3, 20, 55, 16, 65 and 78 find the sequence of servicing using 15

1. Shortest seek time first (SSTF) and 16
2. Elevator disk scheduling policies.

gate1990 descriptive operating-system disk-scheduling

Answer key

5.8.3 Disk Scheduling: GATE CSE 1995 | Question: 20



The head of a moving head disk with 100 tracks numbered 0 to 99 is currently serving a request at track 55. If the queue of requests kept in FIFO order is

10, 70, 75, 23, 65

which of the two disk scheduling algorithms FCFS (First Come First Served) and SSTF (Shortest Seek Time First) will require less head movement? Find the head movement for each of the algorithms.

Answer key

5.8.4 Disk Scheduling: GATE CSE 1997 | Question: 3.6



The correct matching for the following pairs is: 1

(A) Disk Scheduling	(1) Round robin
(B) Batch Processing	(2) SCAN
(C) Time-sharing	(3) LIFO
(D) Interrupt processing	(4) FIFO

2

A. A-3 B-4 C-2 D-1 B. A-4 B-3 C-2 D-1 C. A-2 B-4 C-1 D-3 D. A-3 B-4 C-3 D-2 3

Answer key 5

5.8.5 Disk Scheduling: GATE CSE 1999 | Question: 1.10



Which of the following disk scheduling strategies is likely to give the best throughput? 6

- A. Farthest cylinder next B. Nearest cylinder next 7
C. First come first served D. Elevator algorithm

Answer key 9

5.8.6 Disk Scheduling: GATE CSE 2001 | Question: 20



Consider a disk with the 100 tracks numbered from 0 to 99 rotating at 3000 rpm. The number of sectors per track is 100 and the time to move the head between two successive tracks is 0.2 millisecond. 10

- A. Consider a set of disk requests to read data from tracks 32, 7, 45, 5 and 10. Assuming that the elevator algorithm is used to schedule disk requests, and the head is initially at track 25 moving up (towards larger track numbers), what is the total seek time for servicing the requests? 12
B. Consider an initial set of 100 arbitrary disk requests and assume that no new disk requests arrive while servicing these requests. If the head is initially at track 0 and the elevator algorithm is used to schedule disk requests, what is the worst case time to complete all the requests?

Answer key

5.8.7 Disk Scheduling: GATE CSE 2004 | Question: 12



Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs? 14

- A. 50% B. 40% C. 25% D. 0% 15

Answer key

5.8.8 Disk Scheduling: GATE CSE 2009 | Question: 31



Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

A. 95 ms B. 119 ms C. 233 ms D. 276 ms ¹

gatecse-2009 operating-system disk-scheduling normal

Answer key

5.8.9 Disk Scheduling: GATE CSE 2014 | Set 1 | Question: 19



Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests. ²

gatecse-2014-set1 operating-system disk-scheduling numerical-answers normal ³

Answer key

5.8.10 Disk Scheduling: GATE CSE 2015 | Set 1 | Question: 30



Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: ⁴

45, 20, 90, 10, 50, 60, 80, 25, 70.

Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is _____ tracks. ⁵

gatecse-2015-set1 operating-system disk-scheduling normal numerical-answers ⁶

Answer key

5.8.11 Disk Scheduling: GATE CSE 2016 | Set 1 | Question: 48



Cylinder a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____. ⁷

gatecse-2016-set1 operating-system disk-scheduling normal numerical-answers ⁸

Answer key

5.8.12 Disk Scheduling: GATE CSE 2018 | Question: 53



Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ..., 199), and 256 sectors per track (numbered as 0, 1, ..., 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time: ⁹

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. ¹⁰

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____. ¹¹

gatecse-2018 operating-system disk-scheduling numerical-answers two-marks ¹²

Answer key

5.8.13 Disk Scheduling: GATE CSE 2020 | Question: 35



Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time.

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests. 1

Which one of the following statements is FALSE? 2

- A. T is serviced before P .
 - B. Q is serviced after S , but before T .
 - C. The head reverses its direction of movement between servicing of Q and P .
 - D. R is serviced before P .
- 3

gatecse-2020 operating-system disk-scheduling two-marks 4

Answer key

5.8.14 Disk Scheduling: GATE IT 2004 | Question: 62

A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers.

30 70 115 130 110 80 20 25. 6

How many times will the head change its direction for the disk scheduling policies SSTF (Shortest Seek Time First) and FCFS (First Come First Serve)? 7

- A. 2 and 3
- B. 3 and 3
- C. 3 and 4
- D. 4 and 4 8

gateit-2004 operating-system disk-scheduling normal 9

Answer key

5.8.15 Disk Scheduling: GATE IT 2007 | Question: 82

The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180. 10

Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the requests arrive before the servicing starts? 11

- A. 11, 139, 170, 178, 181, 184, 201, 265
- B. 10, 138, 170, 178, 181, 185, 201, 265 12
- C. 10, 139, 169, 178, 181, 184, 201, 265
- D. 10, 138, 170, 178, 181, 185, 200, 265

gateit-2007 operating-system disk-scheduling normal 13

Answer key

5.8.16 Disk Scheduling: GATE IT 2007 | Question: 83

The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. 14

What is the maximum cardinality of the request set, so that the head changes its direction after servicing every request if the total number of tracks are 2048 and the head can start from any track?

- A. 9
- B. 10
- C. 11
- D. 12 15

gateit-2007 operating-system disk-scheduling normal 16

Answer key 17

5.9 File System (7)

 Practice Test: Test 1 (9Q)

5.9.1 File System: GATE CSE 2002 | Question: 2.22

In the index allocation scheme of blocks to a file, the maximum possible size of the file depends on 18

- A. the size of the blocks, and the size of the address of the blocks.
- B. the number of blocks used for the index, and the size of the blocks.
- C. the size of the blocks, the number of blocks used for the index, and the size of the address of the blocks.
- D. None of the above

Answer key

5.9.2 File System: GATE CSE 2008 | Question: 20



The data blocks of a very large file in the Unix file system are allocated using 1

- A. continuous allocation B. linked allocation 2
C. indexed allocation D. an extension of indexed allocation

gatecse-2008 file-system operating-system normal

Answer key

5.9.3 File System: GATE CSE 2017 | Set 2 | Question: 08



In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed? 8

1. Contiguous 4
2. Linked
3. Indexed

- A. 1 and 3 only B. 2 only C. 3 only D. 2 and 3 only 5

gatecse-2017-set2 operating-system file-system normal 6

Answer key

5.9.4 File System: GATE CSE 2019 | Question: 42



The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) ____ GB 7

gatecse-2019 numerical-answers operating-system file-system two-marks 8

Answer key

5.9.5 File System: GATE CSE 2021 | Set 1 | Question: 15



Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory **foo**. 9

Which of the following operations will necessarily require a full scan of **foo** for successful completion? 10

- A. Creation of a new file in **foo** B. Deletion of an existing file from **foo** 11
C. Renaming of an existing file in **foo** D. Opening of an existing file in **foo**

gatecse-2021-set1 multiple-selects operating-system file-system one-mark 12

Answer key

5.9.6 File System: GATE CSE 2022 | Question: 53



Consider two files systems **A** and **B**, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in **A** and also in **B**. Now, consider inserting a new block in the middle of the file (between 50^{th} and 51^{st} block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in **A** and **B** are n_A and n_B , respectively, then the value of $n_A + n_B$ is _____. 13 14

gatecse-2022 numerical-answers operating-system file-system two-marks 15

Answer key

5.9.7 File System: GATE IT 2004 | Question: 67



In a particular Unix OS, each data block is of size 1024 bytes, each node has 10 direct data block

addresses and three additional addresses: one for single indirect block, one for double indirect block and one for triple indirect block. Also, each block can contain addresses for 128 blocks. Which one of the following is approximately the maximum size of a file in the file system? 1

- A. 512 MB B. 2 GB C. 8 GB D. 16 GB 2

gateit-2004 operating-system file-system normal 3

Answer key 4

5.10

Fork System Call (8)

 **Practice Test:** Test 1 (14Q) 5

5.10.1 Fork System Call: GATE CSE 2005 | Question: 72 6



Consider the following code fragment: 7

```
if (fork() == 0)
{
    a = a + 5;
    printf("%d, %p n", a, &a);
}
else
{
    a = a - 5;
    printf ("%d, %p n", a, & a);
}
```

Let u, v be the values printed by the parent process and x, y be the values printed by the child process. Which one of the following is **TRUE**? 8

- A. $u = x + 10$ and $v = y$ B. $u = x + 10$ and $v! = y$ 9
C. $u + 10 = x$ and $v = y$ D. $u + 10 = x$ and $v! = y$

gatecse-2005 operating-system fork-system-call normal 10

Answer key 11

5.10.2 Fork System Call: GATE CSE 2008 | Question: 66 12



A process executes the following code 12

```
for(i=0; i<n; i++) fork(); 13
```

The total number of child processes created is 14

- A. n B. $2^n - 1$ C. 2^n D. $2^{n+1} - 1$ 15

gatecse-2008 operating-system fork-system-call normal 16

Answer key 17

5.10.3 Fork System Call: GATE CSE 2012 | Question: 8 18



A process executes the code 17

```
fork();
fork();
fork();
```

The total number of child processes created is 19

- A. 3 B. 4 C. 7 D. 8 20

gatecse-2012 operating-system easy fork-system-call 21

Answer key 22

5.10.4 Fork System Call: GATE CSE 2019 | Question: 17 23



The following C program is executed on a Unix/Linux system :

```
#include<unistd.h>
int main()
```

```

{
    int i;
    for(i=0; i<10; i++)
        if(i%2 == 0)
            fork();
    return 0;
}

```

1

The total number of child processes created is _____. 2

gatecse-2019 numerical-answers operating-system fork-system-call one-mark 3

Answer key

5.10.5 Fork System Call: GATE CSE 2023 | Question: 13



Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode? 4

A. Function Call B. malloc Call C. Page Fault D. System Call 5

gatecse-2023 operating-system fork-system-call multiple-selects one-mark 6

Answer key

5.10.6 Fork System Call: GATE CSE 2024 | Set 1 | Question: 47



Consider the following code snippet using the fork () and wait () system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors. 7

```

int x=3;
while (x>0){
    fork ();
    printf("hello");
    wait (NULL) ;
    X-- ;
}

```

8

The total number of times the printf statement is executed is _____. 9

gatecse-2024-set1 numerical-answers operating-system fork-system-call two-marks 10

Answer key

5.10.7 Fork System Call: GATE CSE 2026 | Set 1 | Question: 53



Consider the following program snippet. Assume that the program compiles and runs successfully. Further, assume that the fork() system call is always successful in creating a process. 11

```

int main () {
    int i;
    for (i = 0; i < 3; i++) {
        if (fork() == 0) {
            continue;
        }
        break;
    }
    printf("Hello!");
    return 0;
}

```

12

The total number of times that the printf statement gets executed is _____. (answer in integer) 13

gatecse-2026-set1 numerical-answers operating-system fork-system-call two-marks

Answer key

5.10.8 Fork System Call: GATE IT 2004 | Question: 64



A process executes the following segment of code : 1

```
for(i = 1; i <= n; i++)  
    fork ();
```

2

The number of new processes created is 3

- A. n B. $((n(n+1))/2)$ C. $2^n - 1$ D. $3^n - 1$ 4

gateit-2004 operating-system fork-system-call easy 5

Answer key 6

5.11 7

IO Handling (7) 8

Practice Test: Test 1 (8Q)

9

5.11.1 IO Handling: GATE CSE 1996 | Question: 1.20, ISRO2008-56



10

Which of the following is an example of spooled device? 11

- A. A line printer used to print the output of a number of jobs 12
B. A terminal used to enter input data to a running program
C. A secondary storage device in a virtual memory system
D. A graphic display device

gate1996 operating-system io-handling normal isro2008 13

Answer key

5.11.2 IO Handling: GATE CSE 1998 | Question: 1.29



Which of the following is an example of a spooled device? 14

- A. The terminal used to enter the input data for the C program being executed 15
B. An output device used to print the output of a number of jobs
C. The secondary memory device in a virtual storage system
D. The swapping area on a disk used by the swapper

gate1998 operating-system io-handling easy 16

Answer key

5.11.3 IO Handling: GATE CSE 2005 | Question: 19



Which one of the following is true for a CPU having a single interrupt request line and a single interrupt grant line? 17

- A. Neither vectored interrupt nor multiple interrupting devices are possible 18
B. Vectored interrupts are not possible but multiple interrupting devices are possible
C. Vectored interrupts and multiple interrupting devices are both possible
D. Vectored interrupts are possible but multiple interrupting devices are not possible

gatecse-2005 operating-system io-handling normal 19

Answer key

5.11.4 IO Handling: GATE CSE 2005 | Question: 20



Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instruction privileged. In a

CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O? 1

- A. I/O protection is ensured by operating system routine(s) 2
- B. I/O protection is ensured by a hardware trap
- C. I/O protection is ensured during system configuration
- D. I/O protection is not possible

gatecse-2005 operating-system io-handling normal 3

Answer key

5.11.5 IO Handling: GATE CSE 2018 | Question: 9



The following are some events that occur after a device controller issues an interrupt while process L is under execution. 4

- P. The processor pushes the process status of L onto the control stack 5
- Q. The processor finishes the execution of the current instruction
- R. The processor executes the interrupt service routine
- S. The processor pops the process status of L from the control stack
- T. The processor loads the new PC value based on the interrupt

Which of the following is the correct order in which the events above occur? 6

- A. QPTRS
- B. PTRSQ
- C. TRPQS
- D. QTPRS 7

gatecse-2018 operating-system interrupts normal one-mark co-and-architecture io-handling 8

Answer key

5.11.6 IO Handling: GATE IT 2004 | Question: 11, ISRO2011-33



What is the bit rate of a video terminal unit with 80 characters/line, 8 bits/character and horizontal sweep time of $100\ \mu\text{s}$ (including $20\ \mu\text{s}$ of retrace time)? 9

- A. 8 Mbps
- B. 6.4 Mbps
- C. 0.8 Mbps
- D. 0.64 Mbps 10

gateit-2004 operating-system io-handling easy isro2011 11

Answer key

5.11.7 IO Handling: GATE IT 2006 | Question: 8



Which of the following DMA transfer modes and interrupt handling mechanisms will enable the highest I/O band-width? 12

- A. Transparent DMA and Polling interrupts
- B. Cycle-stealing and Vectored interrupts 14
- C. Block transfer and Vectored interrupts
- D. Block transfer and Polling interrupts

gateit-2006 operating-system io-handling dma normal 15

Answer key

5.12

Input Output (1)

5.12.1 Input Output: GATE CSE 2025 | Set 1 | Question: 19



Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. Answer in integer)

```
int main()
{
```

```
int x=0,i=0;
scanf("%d",&x);
for(i=0; i<20; i++)
{
x = x+20;
printf("%d\n",x);
}
return 0;
}
```

1

gatecse2025-set1 operating-system process-scheduling input-output numerical-answers one-mark 2

Answer key 3

5.13

Inter Process Communication (1)

5.13.1 Inter Process Communication: GATE CSE 1997 | Question: 3.7



4

I/O redirection 5

- A. implies changing the name of a file
- B. can be employed to use an existing file as input file for a program
- C. implies connecting 2 programs through a pipe
- D. None of the above

6

gate1997 operating-system normal inter-process-communication 7

Answer key 8

5.14

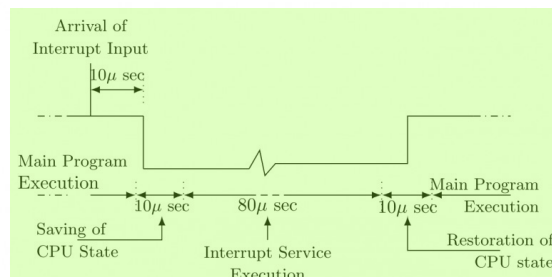
Interrupts (6)

Practice Test: Test 1 (6Q) 9

5.14.1 Interrupts: GATE CSE 1993 | Question: 6.8



The details of an interrupt cycle are shown in figure. 10



11

Given that an interrupt input arrives every 1 msec, what is the percentage of the total time that the CPU devotes for the main program execution. 12

gate1993 operating-system interrupts normal descriptive 13

Answer key

5.14.2 Interrupts: GATE CSE 1997 | Question: 3.8



When an interrupt occurs, an operating system

- A. ignores the interrupt
- B. always changes state of interrupted process after processing the interrupt
- C. always resumes execution of interrupted process after processing the interrupt
- D. may change state of interrupted process to 'blocked' and schedule another process.

gate1997 operating-system interrupts normal

Answer key

5.14.3 Interrupts: GATE CSE 1998 | Question: 1.18



Which of the following devices should get higher priority in assigning interrupts? 1

- A. Hard disk B. Printer C. Keyboard D. Floppy disk 2

gate1998 operating-system interrupts normal 3

Answer key

5.14.4 Interrupts: GATE CSE 1999 | Question: 1.9



Listed below are some operating system abstractions (in the left column) and the hardware components (in the right column) 4

(A) Thread	1. Interrupt
(B) Virtual address space	2. Memory
(C) File system	3. CPU
(D) Signal	4. Disk

 5

- A. (A) – 2 (B) – 4 (C) – 3 (D) – 1 6
B. (A) – 1 (B) – 2 (C) – 3 (D) – 4
C. (A) – 3 (B) – 2 (C) – 4 (D) – 1
D. (A) – 4 (B) – 1 (C) – 2 (D) – 3

gate1999 operating-system easy interrupts virtual-memory disk 7

Answer key

5.14.5 Interrupts: GATE CSE 2001 | Question: 1.12



A processor needs software interrupt to 8

- A. test the interrupt system of the processor 9
B. implement co-routines
C. obtain system services which need execution of privileged instructions
D. return from subroutine

gatecse-2001 operating-system interrupts easy 10

Answer key

5.14.6 Interrupts: GATE CSE 2011 | Question: 11



A computer handles several interrupt sources of which of the following are relevant for this question. 12

- Interrupt from CPU temperature sensor (raises interrupt if CPU temperature is too high) 13
- Interrupt from Mouse (raises Interrupt if the mouse is moved or a button is pressed)
- Interrupt from Keyboard (raises Interrupt if a key is pressed or released)
- Interrupt from Hard Disk (raises Interrupt when a disk read is completed)

Which one of these will be handled at the **HIGHEST** priority? 14

1. Interrupt from Hard Disk 15
2. Interrupt from Mouse
3. Interrupt from Keyboard
4. Interrupt from CPU temperature sensor

gatecse-2011 operating-system interrupts normal

Answer key

5.15.1 Least Recently Used: GATE CSE 2023 | Question: 47



Consider the following two-dimensional array D in the C programming language, which is stored in row-major order:

```
int D[128][128];
```

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D . The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet:

```
for (int i = 0; i < 128; i++)  
    for (int j = 0; j < 128; j++)  
        D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is _____.

gatecse-2023 operating-system page-replacement least-recently-used page-faults numerical-answers two-marks

Answer key

5.16

Linked Allocation (1)

5.16.1 Linked Allocation: GATE CSE 2025 | Set 1 | Question: 41



A disk of size 512 M bytes is divided into blocks of 64 K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1 M bytes needs to be stored in the disk. Assume, $1\text{ K} = 2^{10}$ and $1\text{ M} = 2^{20}$. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)

gatecse2025-set1 operating-system linked-allocation internal-fragmentation numerical-answers two-marks

Answer key

5.17

Memory Management (9)

Practice Tests: Test 1 (15Q) Test 2 (2Q)

5.17.1 Memory Management: GATE CSE 1992 | Question: 12-b



Let the page reference and the working set window be $c c d b c e c e a d$ and 4, respectively. The initial working set at time $t = 0$ contains the pages $\{a, d, e\}$, where a was referenced at time $t = 0$, d was referenced at time $t = -1$, and e was referenced at time $t = -2$. Determine the total number of page faults and the average number of page frames used by computing the working set at each reference.

gate1992 operating-system memory-management normal descriptive

Answer key

5.17.2 Memory Management: GATE CSE 1995 | Question: 5



A computer installation has 1000k of main memory. The jobs arrive and finish in the following sequences.

```
Job 1 requiring 200k arrives  
Job 2 requiring 350k arrives  
Job 3 requiring 300k arrives  
Job 1 finishes  
Job 4 requiring 120k arrives  
Job 5 requiring 150k arrives  
Job 6 requiring 80k arrives
```

- A. Draw the memory allocation table using Best Fit and First Fit algorithms.
B. Which algorithm performs better for this sequence?

gate1995 operating-system memory-management normal descriptive

Answer key

5.17.3 Memory Management: GATE CSE 1996 | Question: 2.18



A 1000 Kbyte memory is managed using variable partitions but no compaction. It currently has two partitions of sizes 200 Kbyte and 260 Kbyte respectively. The smallest allocation request in Kbyte that could be denied is for

- A. 151 B. 181 C. 231 D. 541 ²

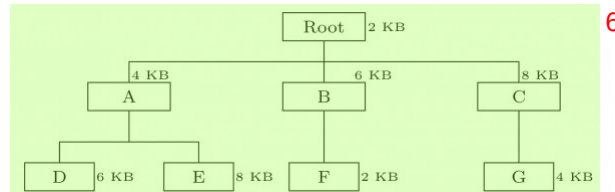
gate1996 operating-system memory-management normal ³

Answer key ⁴

5.17.4 Memory Management: GATE CSE 1998 | Question: 2.16



The overlay tree for a program is as shown below: ⁵



What will be the size of the partition (in physical memory) required to load (and run) this program? ⁷

- A. 12 KB B. 14 KB C. 10 KB D. 8 KB ⁸

gate1998 operating-system normal memory-management ⁹

Answer key ¹⁰

5.17.5 Memory Management: GATE CSE 2014 | Set 2 | Question: 55



Consider the main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____ ¹¹

gatecse-2014-set2 operating-system memory-management numerical-answers normal ¹²

Answer key ¹³

5.17.6 Memory Management: GATE CSE 2015 | Set 2 | Question: 30



Consider 6 memory partitions of sizes 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB, 491 KB in that order. If the best-fit algorithm is used, which partitions are NOT allotted to any process?

- A. 200 KB and 300 KB B. 200 KB and 250 KB ¹⁴
C. 250 KB and 300 KB D. 300 KB and 400 KB

gatecse-2015-set2 operating-system memory-management easy ¹⁵

Answer key ¹⁶

5.17.7 Memory Management: GATE CSE 2020 | Question: 11



Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statement is TRUE?

- A. The hole created by first fit is always larger than the hole created by next fit.
B. The hole created by worst fit is always larger than the hole created by first fit.
C. The hole created by best fit is never larger than the hole created by first fit.
D. The hole created by next fit is never larger than the hole created by best fit.

gatecse-2020 operating-system memory-management one-mark

Answer key

5.17.8 Memory Management: GATE IT 2006 | Question: 56

For each of the four processes P_1, P_2, P_3 , and P_4 . The total size in kilobytes (KB) and the number of segments are given below.

Process	Total size (in KB)	Number of segments
P_1	195	4
P_2	254	5
P_3	45	3
P_4	364	8

The page size is 1 KB. The size of an entry in the page table is 4 bytes. The size of an entry in the segment table is 8 bytes. The maximum size of a segment is 256 KB. The paging method for memory management uses two-level paging, and its storage overhead is P . The storage overhead for the segmentation method is S . The storage overhead for the segmentation and paging method is T . What is the relation among the overheads for the different methods of memory management in the concurrent execution of the above four processes?

- A. $P < S < T$
B. $S < P < T$
C. $S < T < P$
D. $T < S < P$

gateit-2006 operating-system memory-management difficult

Answer key

5.17.9 Memory Management: GATE IT 2007 | Question: 11

Let a memory have four free blocks of sizes $4k, 8k, 20k, 2k$. These blocks are allocated following the best-fit strategy. The allocation requests are stored in a queue as shown below.

Request No	J1	J2	J3	J4	J5	J6	J7	J8
Request Sizes	2k	14k	3k	6k	6k	10k	7k	20k
Usage Time	4	10	2	8	4	1	8	6

The time at which the request for $J7$ will be completed will be

- A. 16
B. 19
C. 20
D. 37

gateit-2007 operating-system memory-management normal

Answer key

5.18 Multilevel Paging (1)

Practice Test: Test 1 (10Q)

5.18.1 Multilevel Paging: GATE CSE 2025 | Set 2 | Question: 48

A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b -bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size.

The value of b is _____. (Answer in integer)

gatecse2025-set2 operating-system paging multilevel-paging numerical-answers two-marks

Answer key

5.19 OS Protection (3)

5.19.1 OS Protection: GATE CSE 1999 | Question: 1.11, UGCNET-Dec2015-II: 44

System calls are usually invoked by using